US012412481B2

(12) **United States Patent**
Hammond

(10) **Patent No.: US 12,412,481 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **CONTEXT-BASED NAVIGATION OF UNCREWED VEHICLES USING RELATIVE POSITION MARKERS**

(71) Applicant: **Wing Aviation LLC**, Mountain View, CA (US)

(72) Inventor: **Marcus Hammond**, Redwood City, CA (US)

(73) Assignee: **Wing Aviation LLC**, Palo Alto, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 264 days.

(21) Appl. No.: **18/077,371**

(22) Filed: **Dec. 8, 2022**

(65) **Prior Publication Data**

US 2024/0194081 A1 Jun. 13, 2024

(51) **Int. Cl.**

G08G 5/00	(2025.01)
B64C 39/02	(2023.01)
B64U 10/14	(2023.01)
B64U 50/37	(2023.01)
B64U 70/95	(2023.01)

(Continued)

(52) **U.S. Cl.**

CPC ***G08G 5/55*** (2025.01); ***B64U 10/14*** (2023.01); ***B64U 50/37*** (2023.01); ***B64U 70/95*** (2023.01); ***G06V 20/17*** (2022.01); ***G08G 5/57*** (2025.01); *B64U 50/34* (2023.01); *B64U 70/00* (2023.01)

(58) **Field of Classification Search**

CPC ... G08G 5/55; G08G 5/57; G08G 5/56; B64U 10/14; B64U 50/37; B64U 70/95; B64U 50/34; B64U 70/00; B64U 2201/10; B64U 2101/30; G06V 20/17; G06V 20/176; G06V 20/13; H10H 29/922; H10D 84/0126; H10F 77/1692; B64C 39/024

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,223,666 B1	3/2019	Palamarchuk et al.
11,287,835 B2	3/2022	Jourdan et al.

(Continued)

OTHER PUBLICATIONS

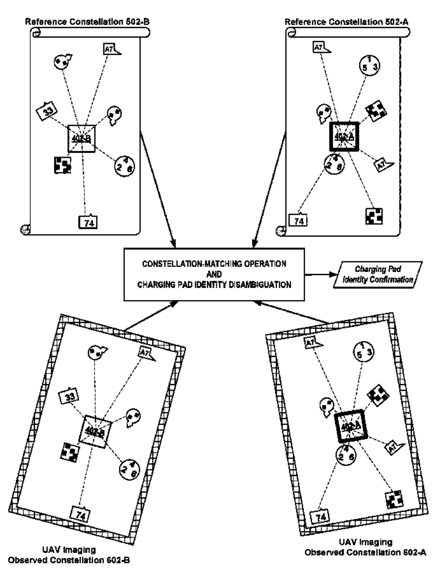
International Searching Authority, International Search Report and Written Opinion mailed Mar. 6, 2024, issued in connection with International Patent Application No. PCT/US2023/078831, filed Nov. 6, 2023, 13 pages.

Primary Examiner — Sahar Motazedi
(74) *Attorney, Agent, or Firm* — McDonnell Boehnen Hulbert & Berghoff LLP

(57) **ABSTRACT**

In an example embodiment, a method carried out by an uncrewed aerial vehicle (UAV) may involve receiving a reference map of a cluster of charging pads from a server. The cluster may include a layout of charging pads and fiducial markers distributed across the layout, the reference map representing the layout and fiducial markers. The UAV may fly to the cluster and acquire an image of charging pads and observed fiducial markers near the charging pads. The image may capture an observed constellation of fiducial markers at apparent positions and orientations relative to the charging pads. A reference constellation of fiducial markers at reference positions and orientations relative to reference charging pads may be identified in the reference map. Identities of the reference charging pads and a match of the reference constellation to the observed constellation may be used to disambiguate a particular charging pad from among the charging pads.

20 Claims, 14 Drawing Sheets



(51) **Int. Cl.**

G06V 20/17	(2022.01)
G08G 5/55	(2025.01)
G08G 5/57	(2025.01)
B64U 50/34	(2023.01)
B64U 70/00	(2023.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,521,500	B1 *	12/2022	Parodi	G01S 15/10
2005/0090972	A1	4/2005	Bodin et al.	
2015/0353206	A1	12/2015	Wang	
2016/0039542	A1	2/2016	Wang	
2016/0091899	A1 *	3/2016	Aldred	B25J 5/00 901/1
2016/0122038	A1	5/2016	Fleischman et al.	
2017/0011333	A1	1/2017	Greiner et al.	
2017/0031369	A1	2/2017	Liu et al.	
2017/0323129	A1	11/2017	Davidson et al.	
2018/0053139	A1	2/2018	Stoman	
2019/0197908	A1	6/2019	Mozer et al.	
2019/0278288	A1 *	9/2019	Xiong	G05D 1/0274
2020/0130864	A1	4/2020	Brockers et al.	
2020/0165007	A1	5/2020	Augugliaro et al.	
2020/0225357	A1	7/2020	Smits	
2020/0301445	A1 *	9/2020	Jourdan	G06K 19/06037
2021/0284356	A1 *	9/2021	Jourdan	G06V 10/245
2023/0356863	A1 *	11/2023	Zhang	G06T 7/75
2024/0192705	A1 *	6/2024	Reddy	G06T 7/73

* cited by examiner

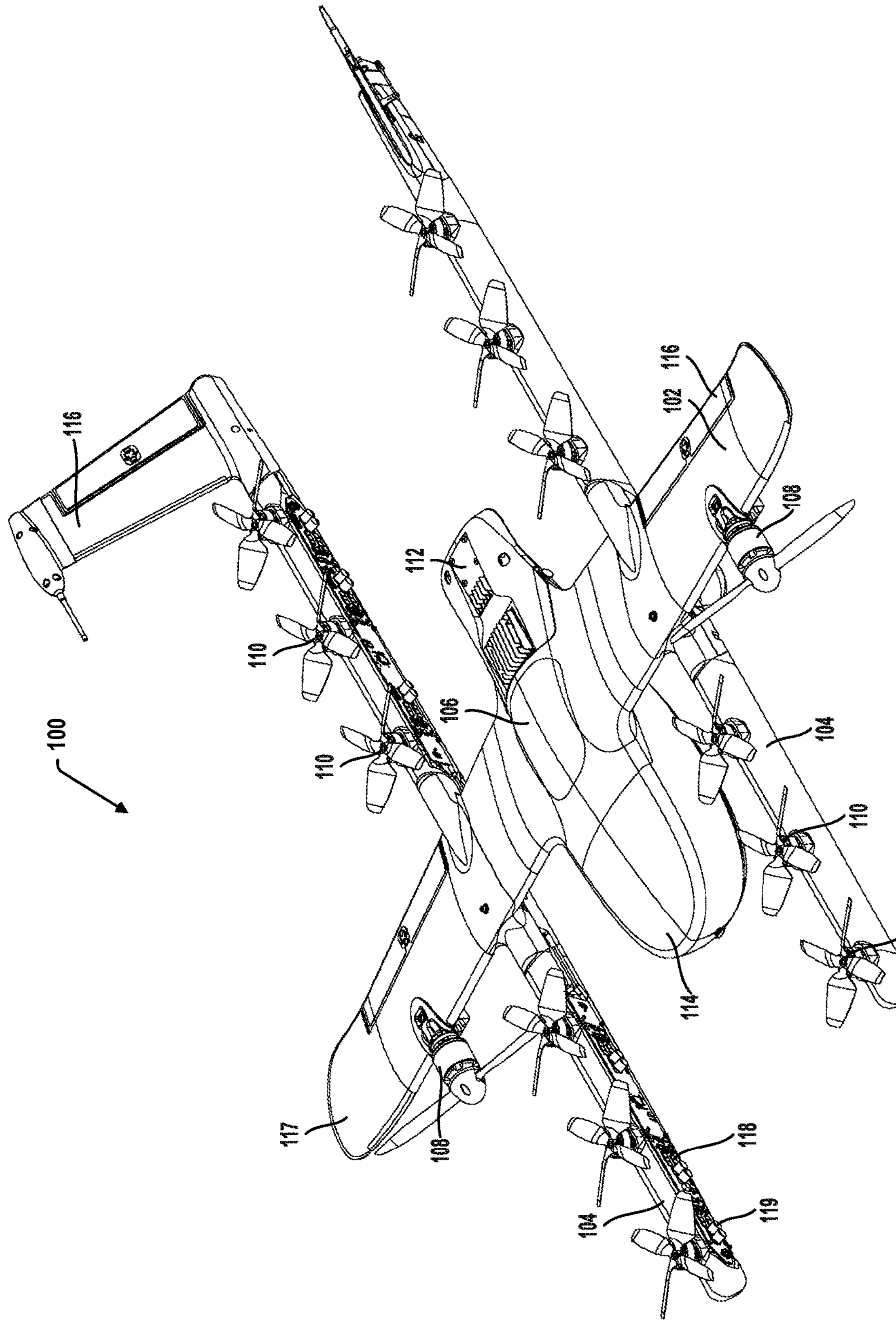

100
116
116
102
108
112
110
110
106
104
110
110
114
117
108
118
104
119

Figure 1A

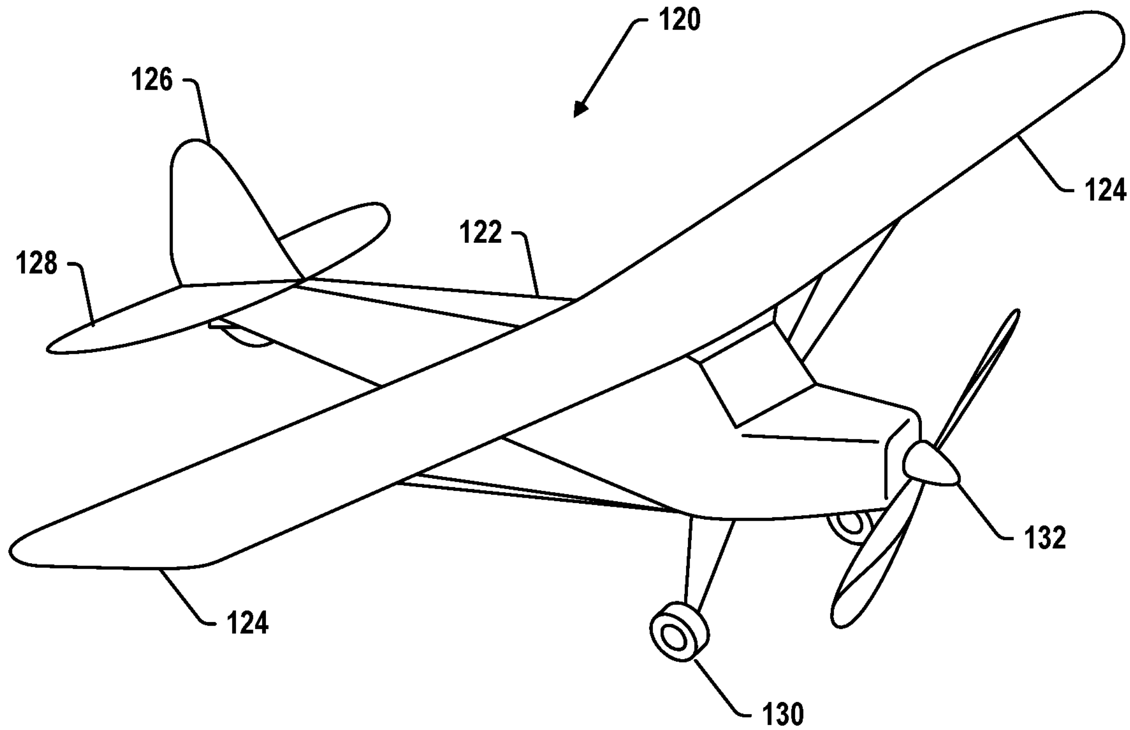


Figure 1B

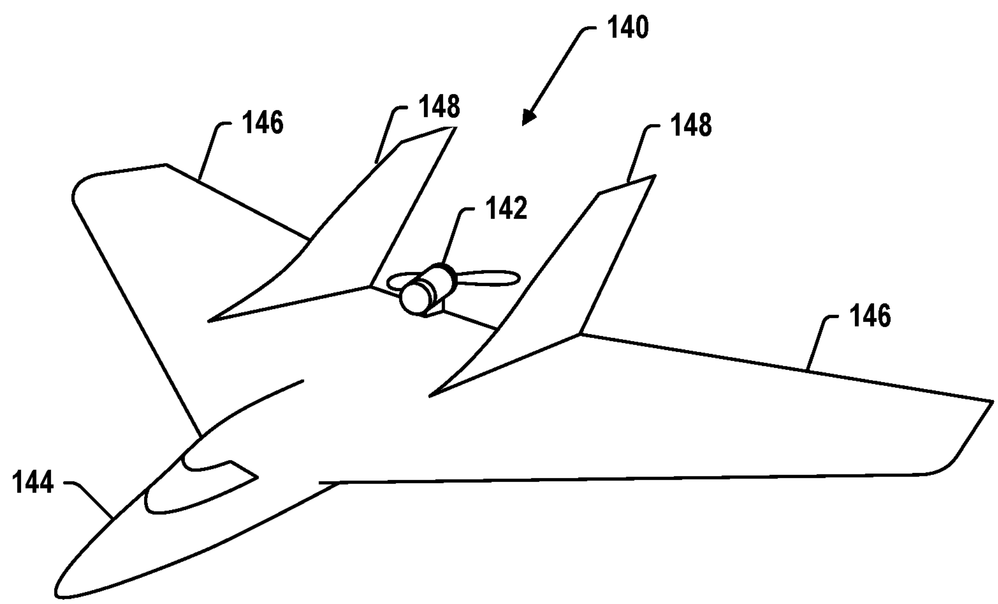


Figure 1C

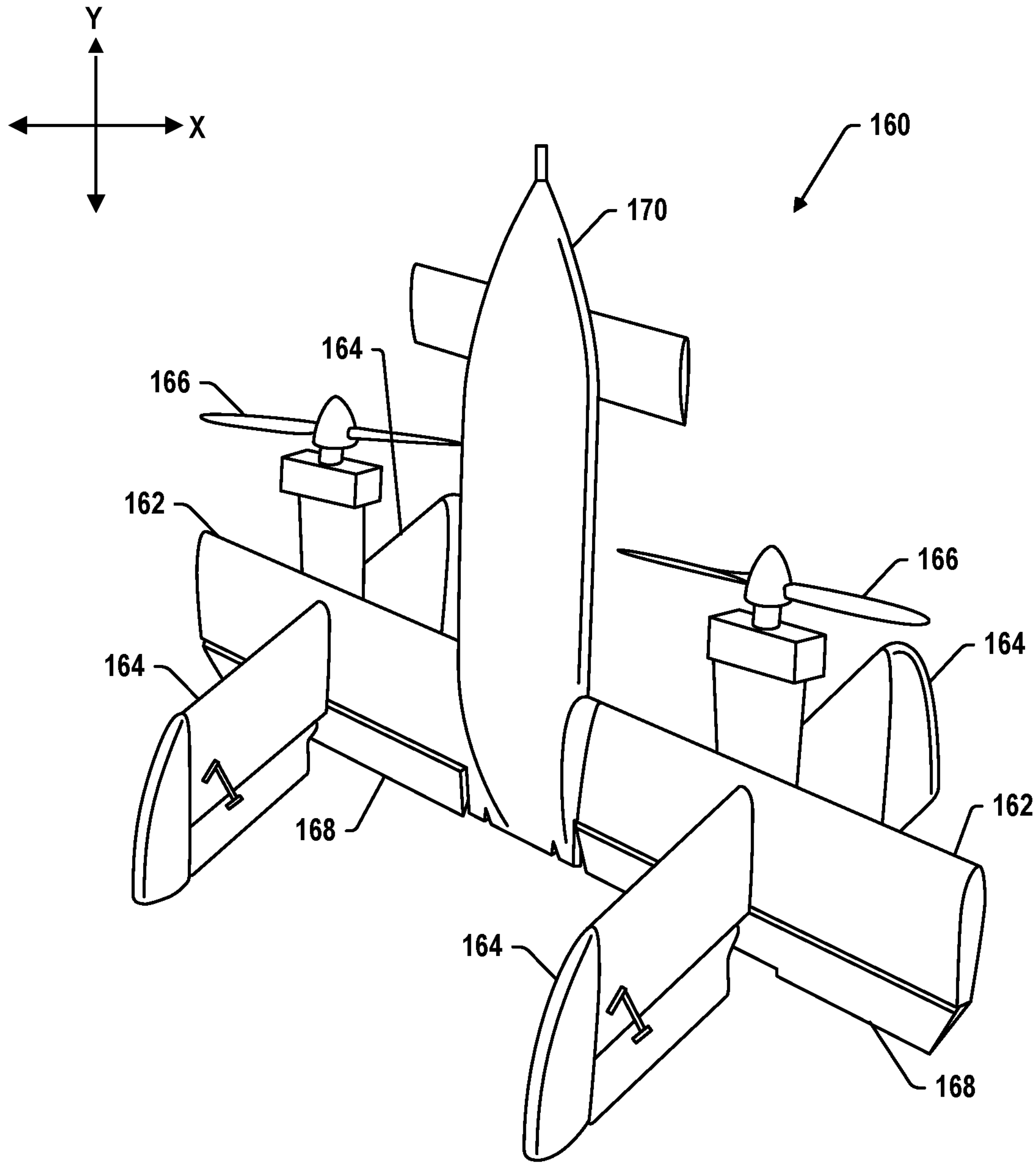


Figure 1D

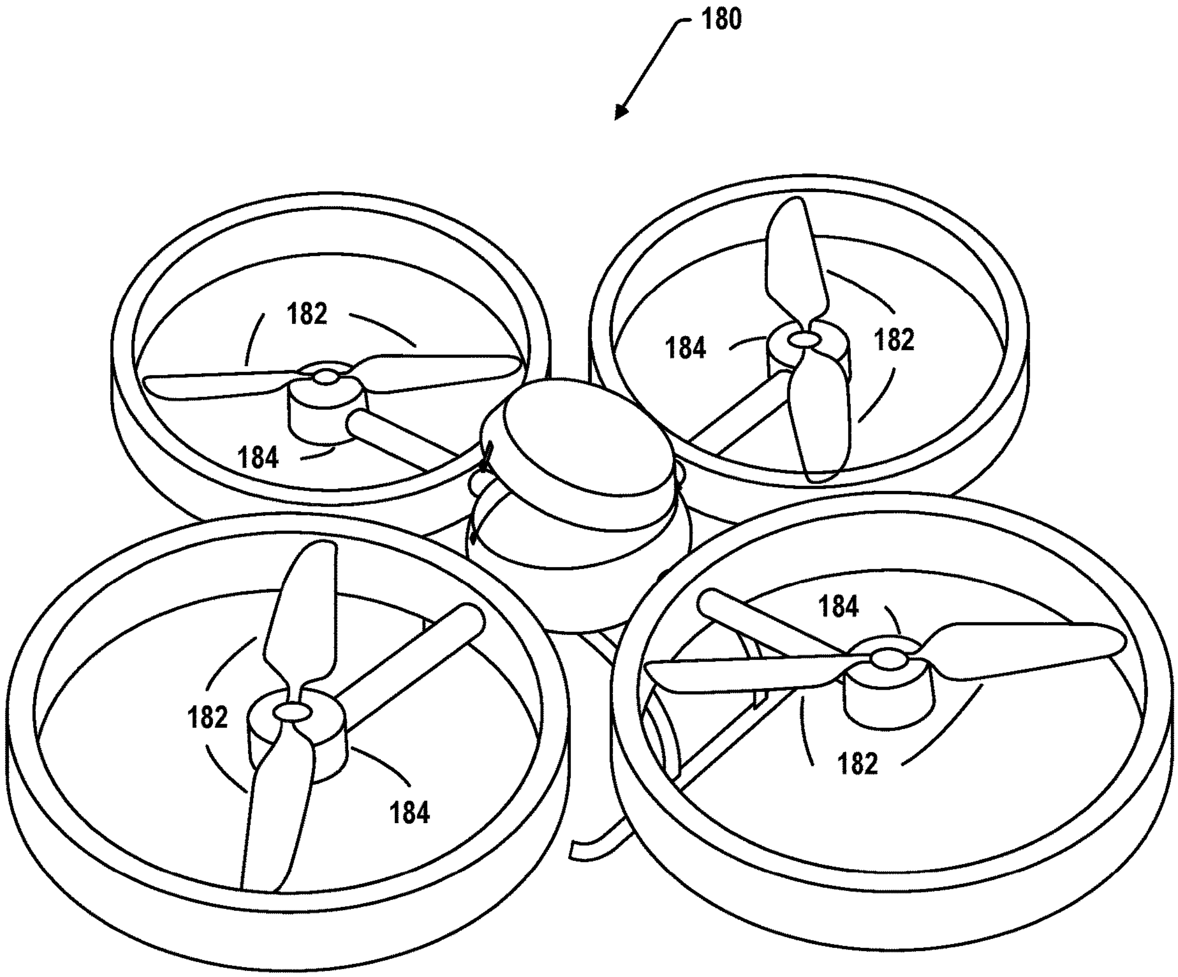


FIG. 1E

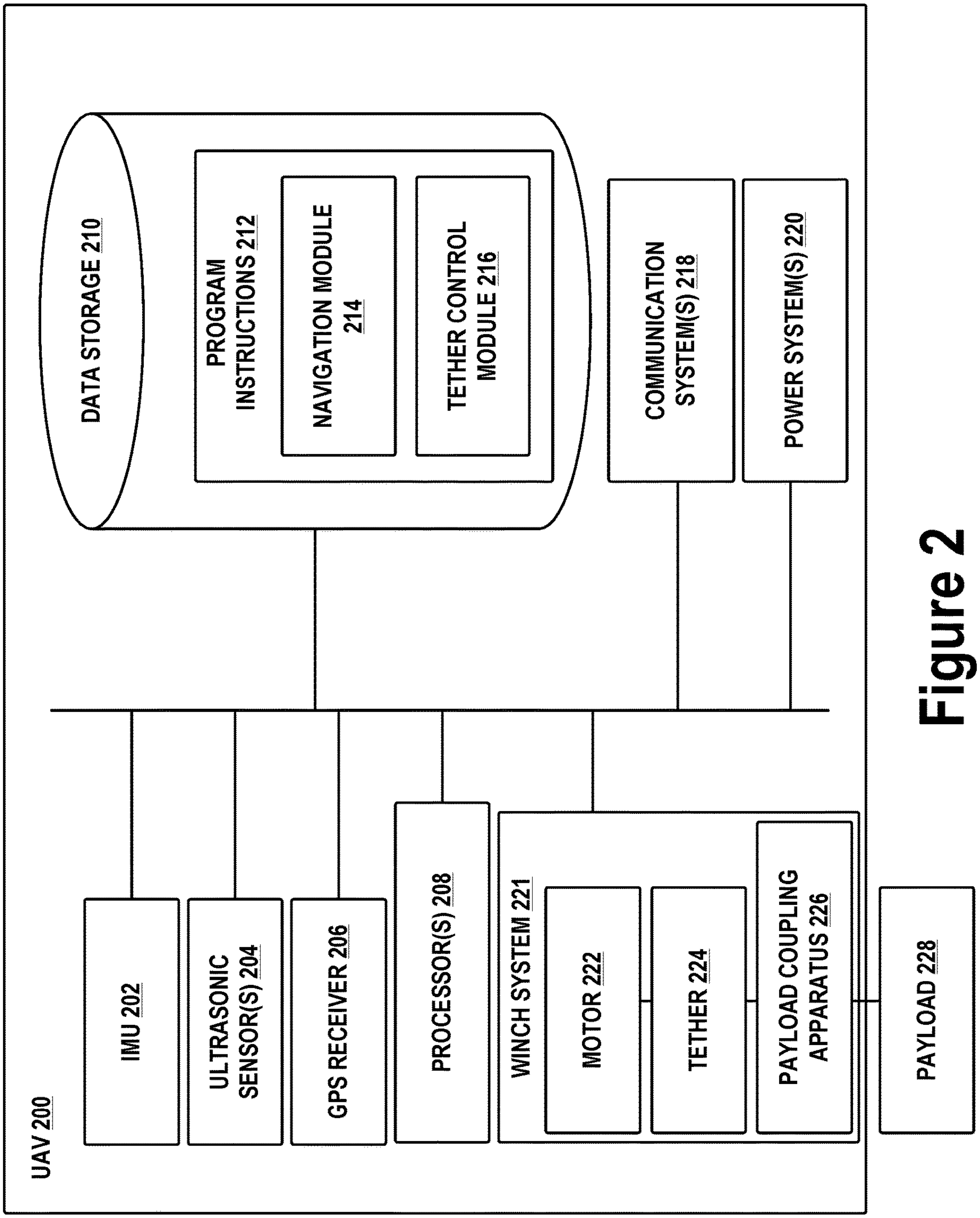

UAV 200
IMU 202
ULTRASONIC SENSOR(S) 204
GPS RECEIVER 206
PROCESSOR(S) 208
DATA STORAGE 210
PROGRAM INSTRUCTIONS 212
NAVIGATION MODULE 214
TETHER CONTROL MODULE 216
COMMUNICATION SYSTEM(S) 218
POWER SYSTEM(S) 220
WINCH SYSTEM 221
MOTOR 222
TETHER 224
PAYLOAD COUPLING APPARATUS 226
PAYLOAD 228

Figure 2

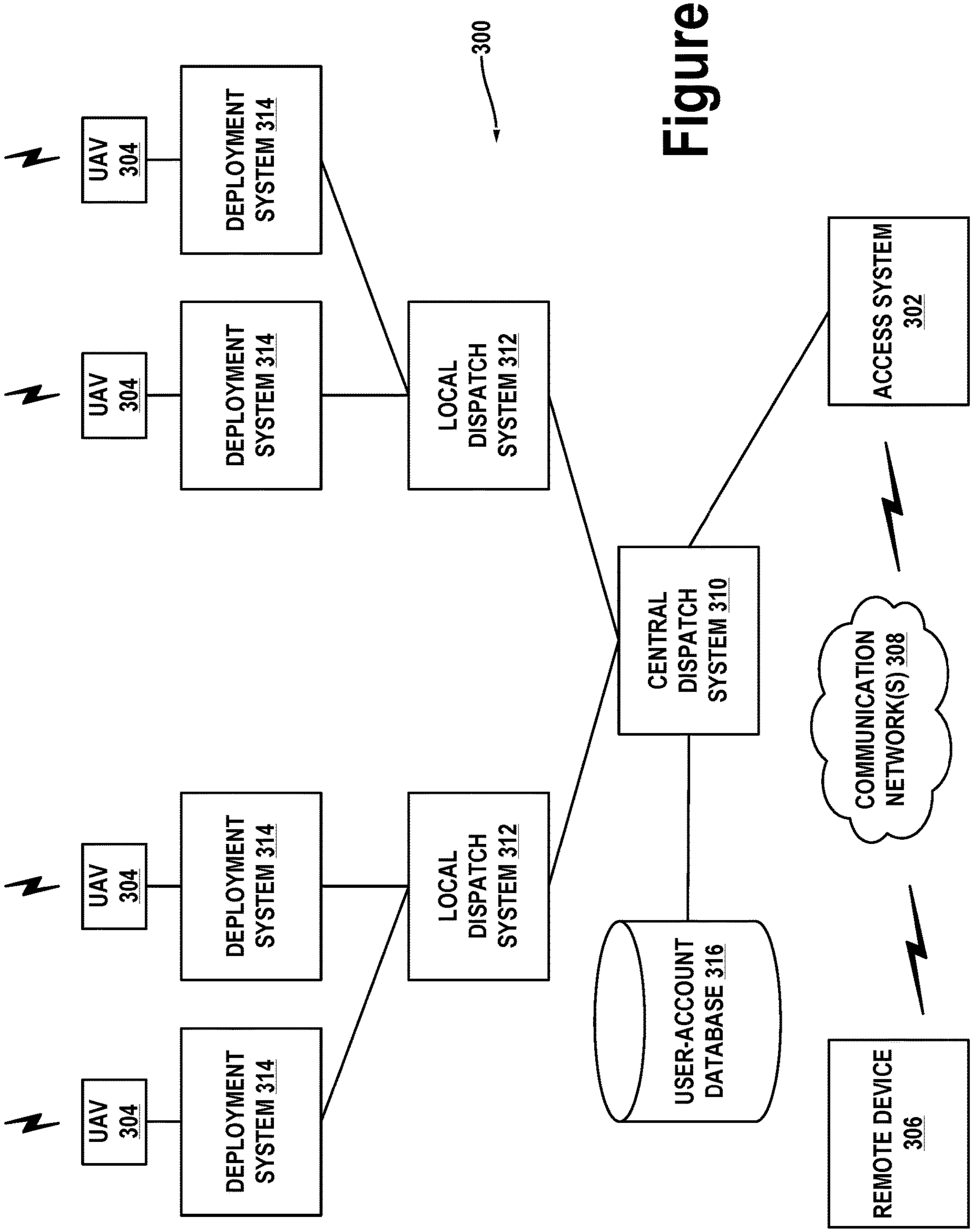

300
UAV 304
DEPLOYMENT SYSTEM 314
UAV 304
DEPLOYMENT SYSTEM 314
LOCAL DISPATCH SYSTEM 312
UAV 304
DEPLOYMENT SYSTEM 314
UAV 304
DEPLOYMENT SYSTEM 314
LOCAL DISPATCH SYSTEM 312
CENTRAL DISPATCH SYSTEM 310
ACCESS SYSTEM 302
COMMUNICATION NETWORK(S) 308
USER-ACCOUNT DATABASE 316
REMOTE DEVICE 306

Figure 3

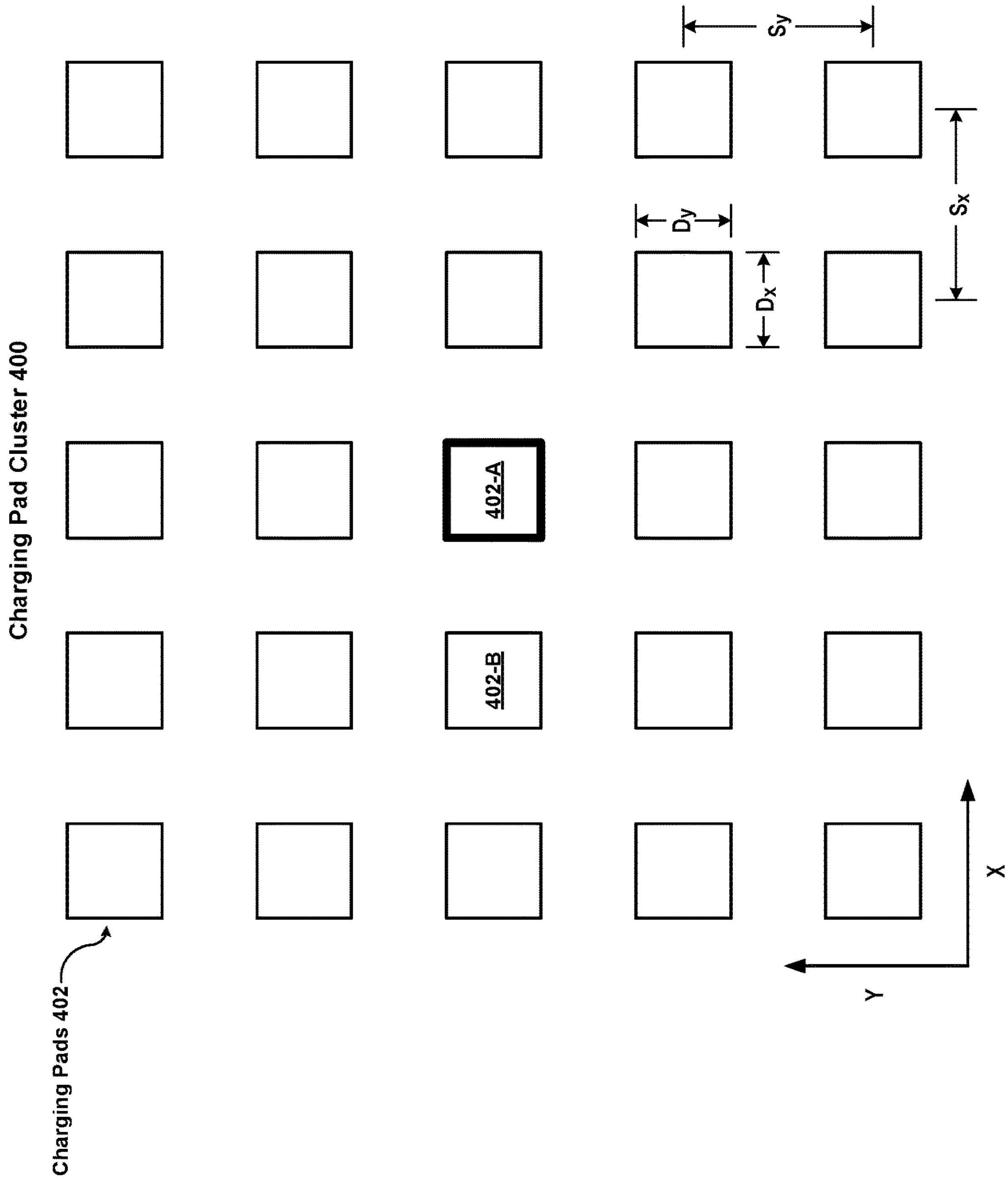

Charging Pad Cluster 400
Charging Pads 402
402-A
402-B
Sy
Sx
Dy
Dx
X
Y

Figure 4A

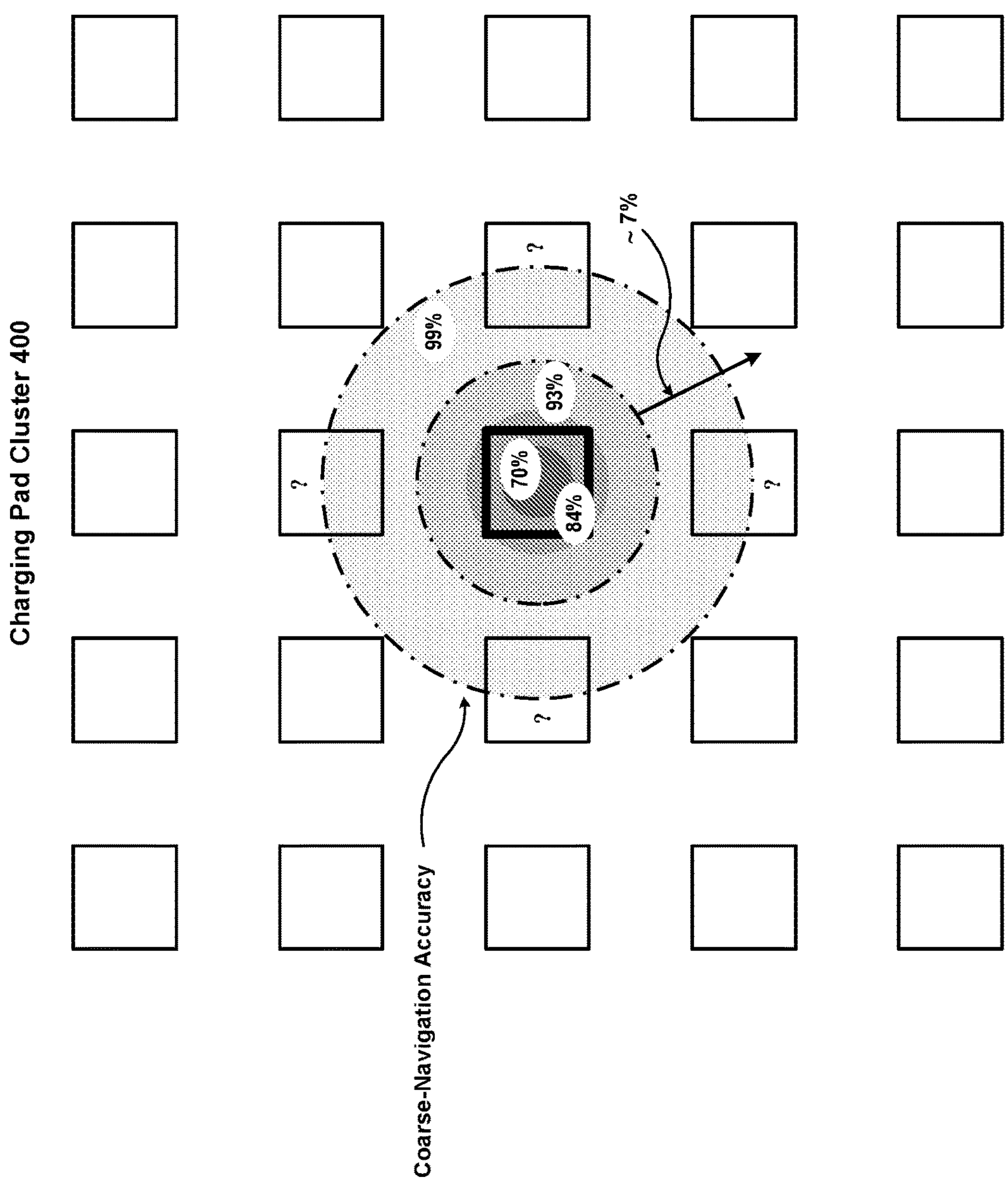

Charging Pad Cluster 400
~ 7%
99%
93%
70%
84%
?
?
?
?
Coarse-Navigation Accuracy

Figure 4B

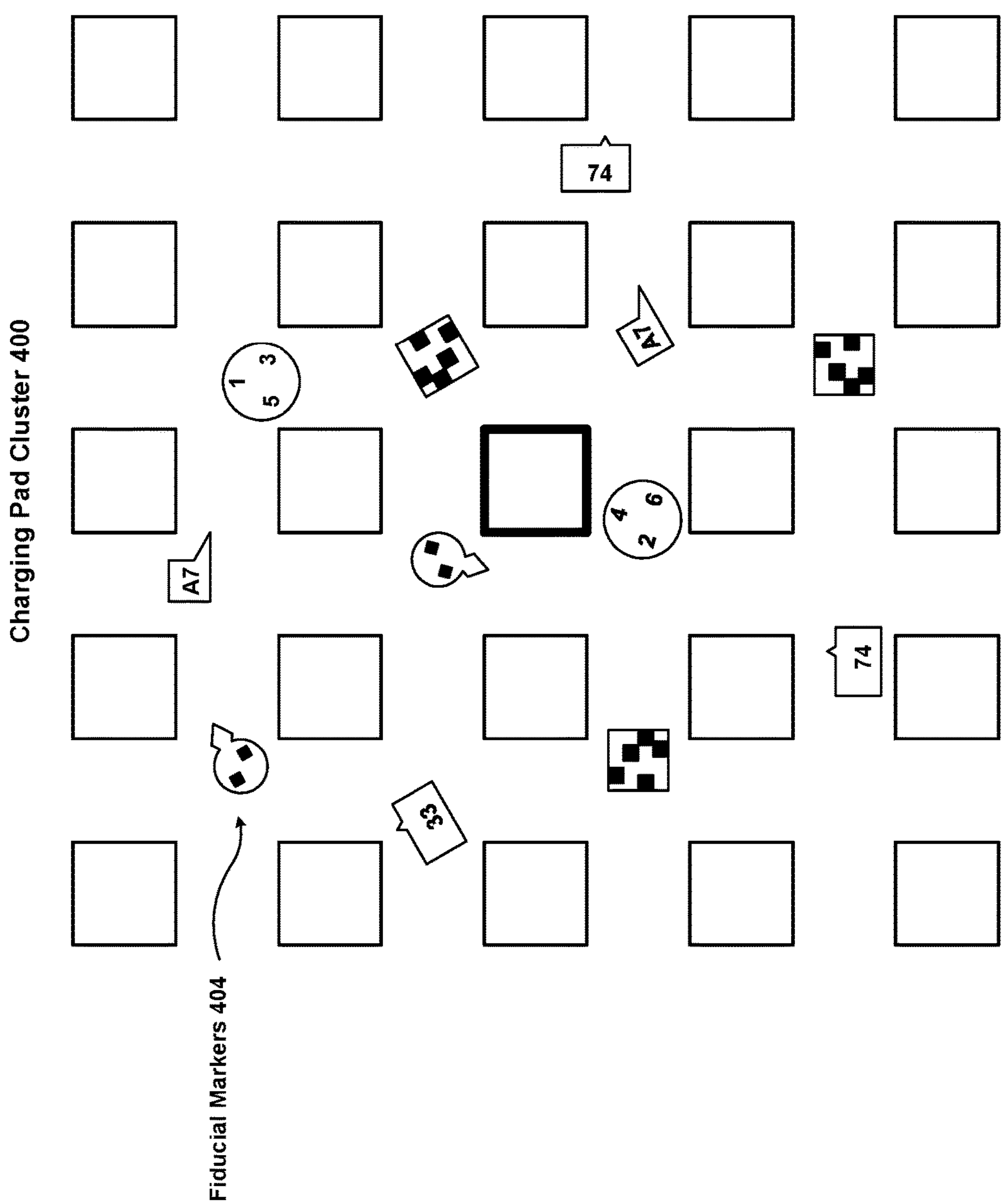

Charging Pad Cluster 400
Fiducial Markers 404
74
A7
33
1 3 5
4 6 2

Figure 4C

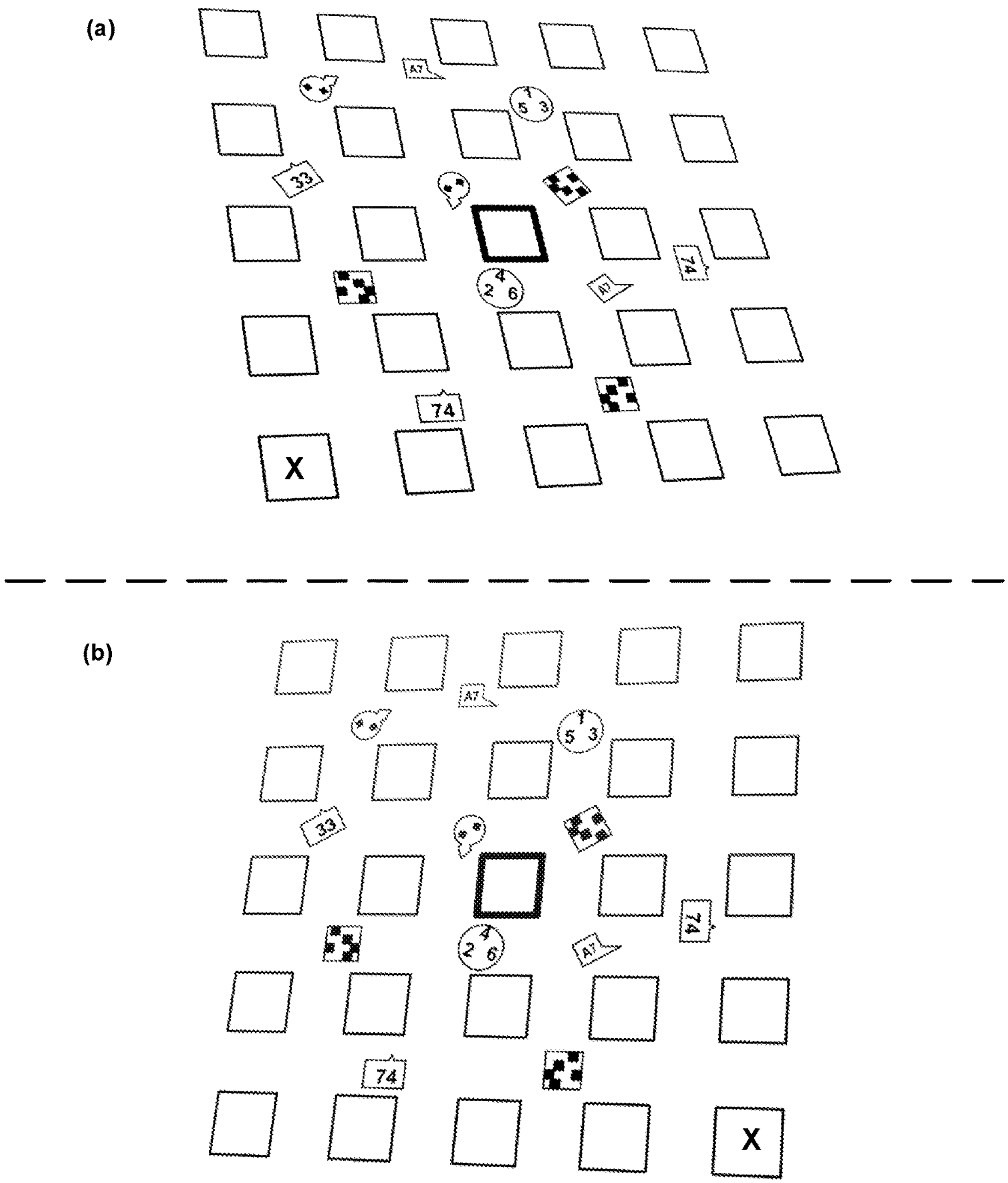


Figure 4D

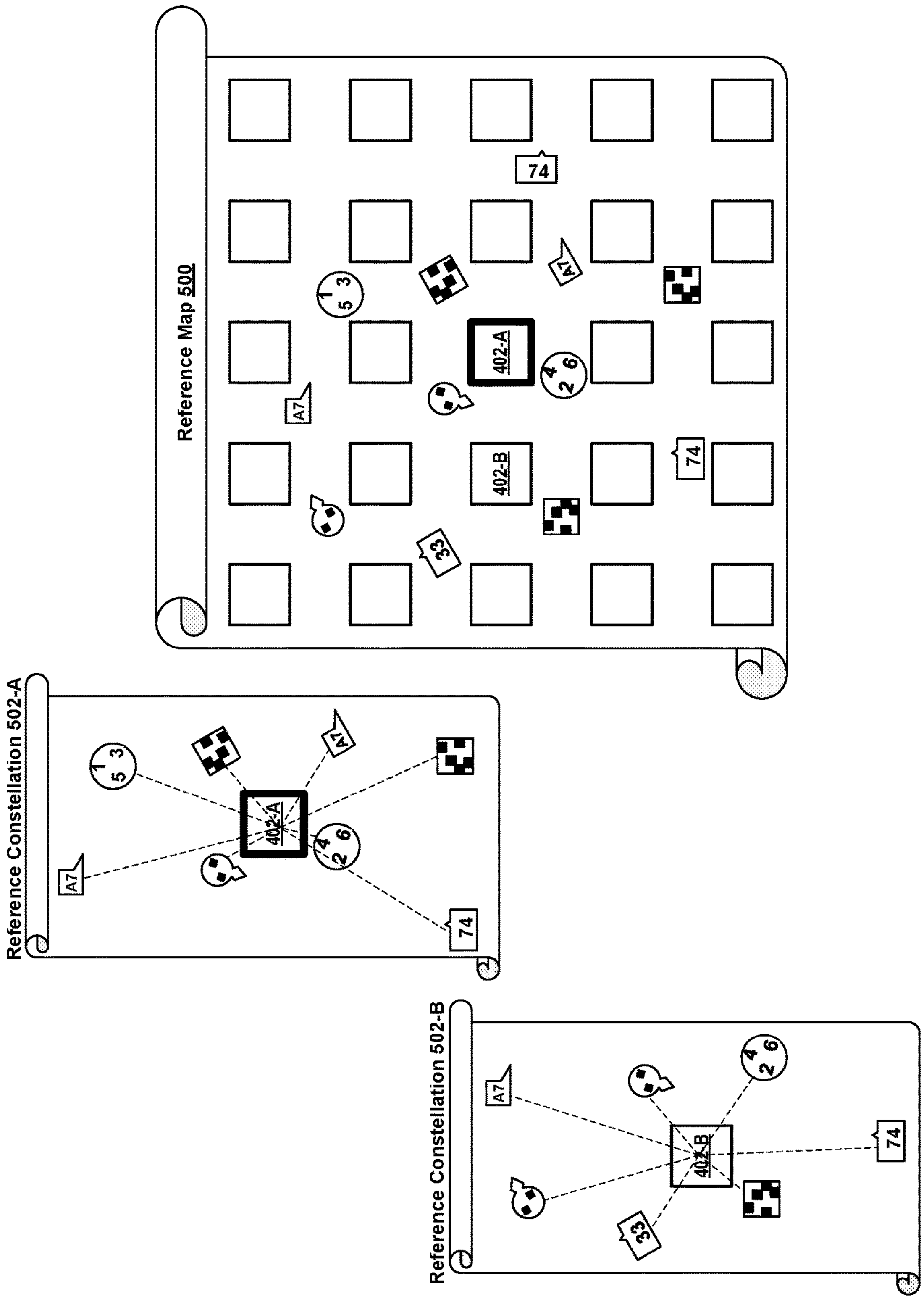

Reference Map 500
402-A
402-B
74
A7
33
Reference Constellation 502-A
Reference Constellation 502-B

Figure 5

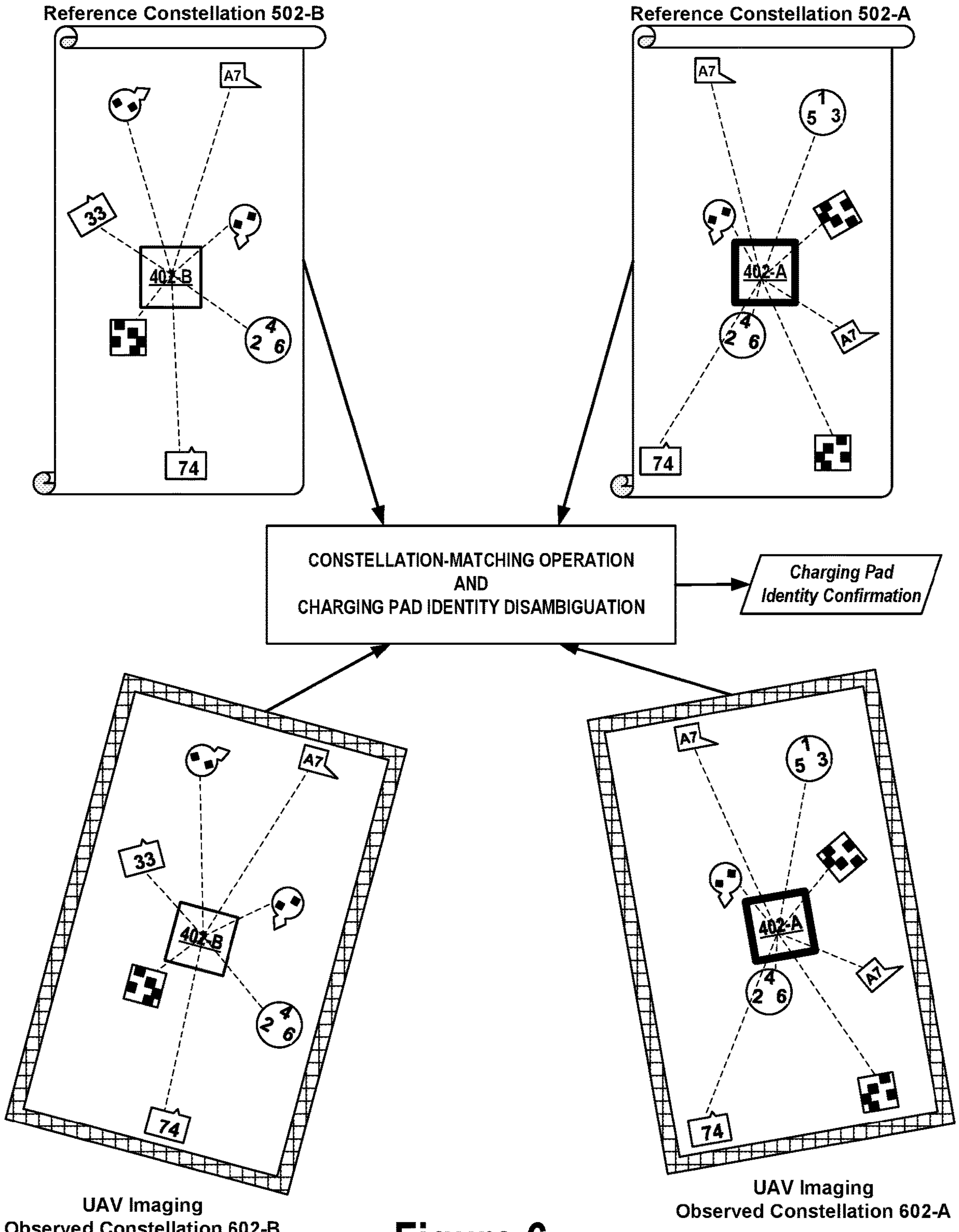

Reference Constellation 502-B
A7
33
402-B
2 4 6
74
Reference Constellation 502-A
A7
1 5 3
402-A
A7
4 2 6
74
CONSTELLATION-MATCHING OPERATION
AND
CHARGING PAD IDENTITY DISAMBIGUATION
Charging Pad
Identity Confirmation
A7
33
402-B
4 2 6
74
A7
1 5 3
402-A
A7
4 2 6
74
UAV Imaging
Observed Constellation 602-B
UAV Imaging
Observed Constellation 602-A

Figure 6

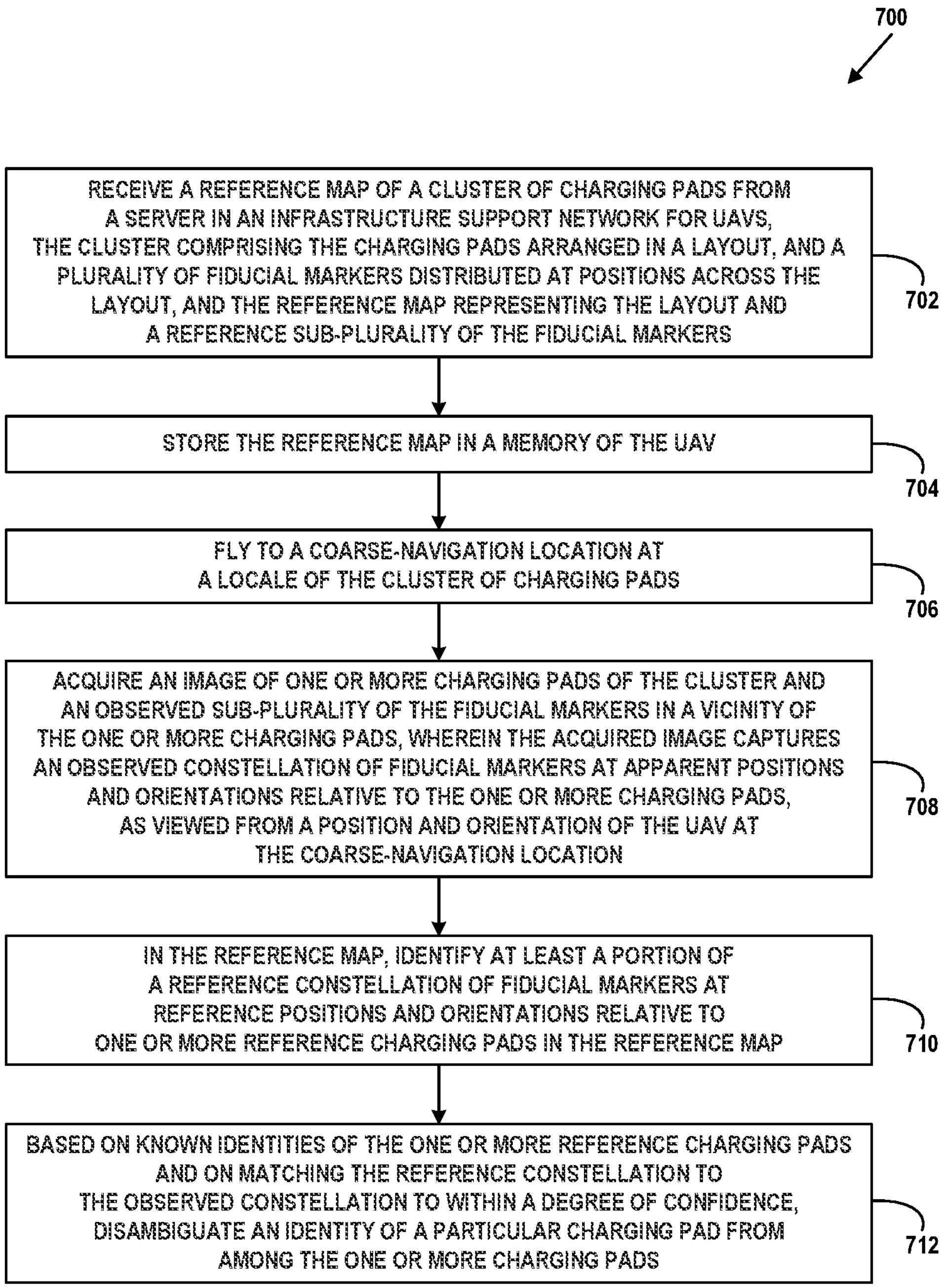

700
RECEIVE A REFERENCE MAP OF A CLUSTER OF CHARGING PADS FROM A SERVER IN AN INFRASTRUCTURE SUPPORT NETWORK FOR UAVS, THE CLUSTER COMPRISING THE CHARGING PADS ARRANGED IN A LAYOUT, AND A PLURALITY OF FIDUCIAL MARKERS DISTRIBUTED AT POSITIONS ACROSS THE LAYOUT, AND THE REFERENCE MAP REPRESENTING THE LAYOUT AND A REFERENCE SUB-PLURALITY OF THE FIDUCIAL MARKERS
702
STORE THE REFERENCE MAP IN A MEMORY OF THE UAV
704
FLY TO A COARSE-NAVIGATION LOCATION AT A LOCALE OF THE CLUSTER OF CHARGING PADS
706
ACQUIRE AN IMAGE OF ONE OR MORE CHARGING PADS OF THE CLUSTER AND AN OBSERVED SUB-PLURALITY OF THE FIDUCIAL MARKERS IN A VICINITY OF THE ONE OR MORE CHARGING PADS, WHEREIN THE ACQUIRED IMAGE CAPTURES AN OBSERVED CONSTELLATION OF FIDUCIAL MARKERS AT APPARENT POSITIONS AND ORIENTATIONS RELATIVE TO THE ONE OR MORE CHARGING PADS, AS VIEWED FROM A POSITION AND ORIENTATION OF THE UAV AT THE COARSE-NAVIGATION LOCATION
708
IN THE REFERENCE MAP, IDENTIFY AT LEAST A PORTION OF A REFERENCE CONSTELLATION OF FIDUCIAL MARKERS AT REFERENCE POSITIONS AND ORIENTATIONS RELATIVE TO ONE OR MORE REFERENCE CHARGING PADS IN THE REFERENCE MAP
710
BASED ON KNOWN IDENTITIES OF THE ONE OR MORE REFERENCE CHARGING PADS AND ON MATCHING THE REFERENCE CONSTELLATION TO THE OBSERVED CONSTELLATION TO WITHIN A DEGREE OF CONFIDENCE, DISAMBIGUATE AN IDENTITY OF A PARTICULAR CHARGING PAD FROM AMONG THE ONE OR MORE CHARGING PADS
712

Figure 7

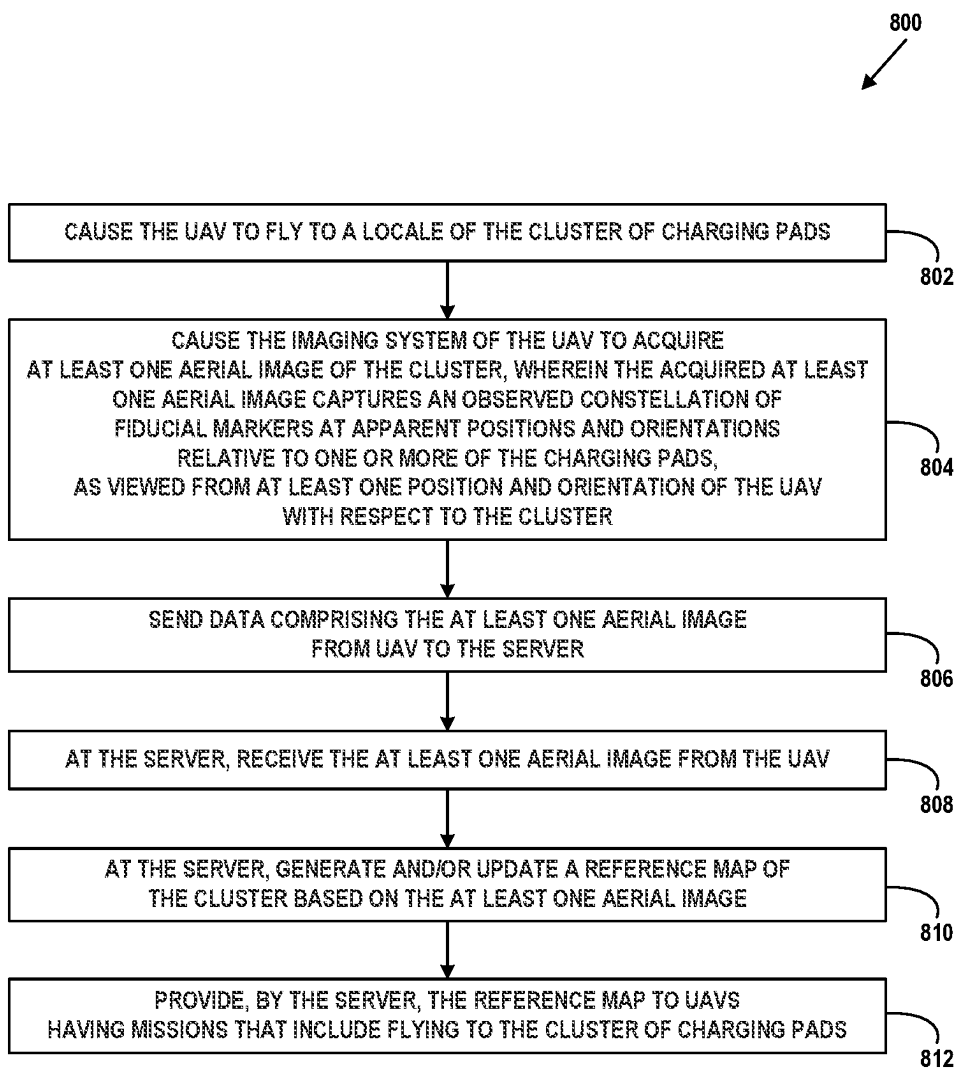

800
CAUSE THE UAV TO FLY TO A LOCALE OF THE CLUSTER OF CHARGING PADS
802
CAUSE THE IMAGING SYSTEM OF THE UAV TO ACQUIRE AT LEAST ONE AERIAL IMAGE OF THE CLUSTER, WHEREIN THE ACQUIRED AT LEAST ONE AERIAL IMAGE CAPTURES AN OBSERVED CONSTELLATION OF FIDUCIAL MARKERS AT APPARENT POSITIONS AND ORIENTATIONS RELATIVE TO ONE OR MORE OF THE CHARGING PADS, AS VIEWED FROM AT LEAST ONE POSITION AND ORIENTATION OF THE UAV WITH RESPECT TO THE CLUSTER
804
SEND DATA COMPRISING THE AT LEAST ONE AERIAL IMAGE FROM UAV TO THE SERVER
806
AT THE SERVER, RECEIVE THE AT LEAST ONE AERIAL IMAGE FROM THE UAV
808
AT THE SERVER, GENERATE AND/OR UPDATE A REFERENCE MAP OF THE CLUSTER BASED ON THE AT LEAST ONE AERIAL IMAGE
810
PROVIDE, BY THE SERVER, THE REFERENCE MAP TO UAVS HAVING MISSIONS THAT INCLUDE FLYING TO THE CLUSTER OF CHARGING PADS
812

Figure 8

CONTEXT-BASED NAVIGATION OF UNCREWED VEHICLES USING RELATIVE POSITION MARKERS

BACKGROUND

An uncrewed vehicle, which may also be referred to as an autonomous vehicle, is a vehicle capable of travel without a physically-present human operator. An uncrewed vehicle may operate in a remote-control mode, in an autonomous mode, or in a partially autonomous mode. The term “unmanned” may sometimes be used instead of, or in addition to, “uncrewed,” and for the purposes of this disclosure, it should be understood that both terms have the same meaning, and may be used interchangeably.

When an uncrewed vehicle operates in a remote-control mode, a pilot or driver that is at a remote location can control the uncrewed vehicle via commands that are sent to the uncrewed vehicle via a wireless link. When the uncrewed vehicle operates in autonomous mode, the uncrewed vehicle typically moves based on pre-programmed navigation waypoints, dynamic automation systems, or a combination of these. Further, some uncrewed vehicles can operate in both a remote-control mode and an autonomous mode, and in some instances may do so simultaneously. For instance, a remote pilot or driver may wish to leave navigation to an autonomous system while manually performing another task, such as operating a mechanical system for picking up objects, as an example.

Various types of uncrewed vehicles exist for various different environments. For instance, uncrewed vehicles exist for operation in the air, on the ground, underwater, and in space. Examples include quad-copters and tail-sitter uncrewed aerial vehicles (UAVs), among others. Uncrewed vehicles also exist for hybrid operations in which multi-environment operation is possible. Examples of hybrid uncrewed vehicles include an amphibious craft that is capable of operation on land as well as on water or a floatplane that is capable of landing on water as well as on land. Other examples are also possible.

SUMMARY

Examples disclosed herein include systems and methods involving navigation of UAVs based, at least in part, on fiducial markers placed at various positions across an area containing navigational targets for UAVs. Navigational targets may include a group or cluster of charging pads to which UAVs intend, or are instructed, to navigate, based on mission specifications, for example, and upon designated charging pads of which UAVs may further be directed to land, based on operational procedures, for example. More particularly, as part of mission operations or protocols, UAVs may, from time to time and/or according to mission plans, land on designated charging pads to charge or recharge their batteries (among other possible reasons for landing). A given UAV may travel (fly) to a locale of a cluster of charging pads arranged in layout, such as a grid, navigate to a specific one of the charging pads designated as a landing pad, and then land on the designated pad. In accordance with example embodiments, a plurality of fiducial markers may be distributed across a layout of charging pads, and patterns or constellations that the fiducial markers form at positions and orientations relative to the charging pads as viewed from above by UAVs may be used by the UAVs to disambiguate or distinguish designated target charging pads from among the cluster.

In a first aspect, an uncrewed aerial vehicle (UAV) may comprise a navigation system, an imaging system, and a control system. The control system may be configured to: cause the UAV to fly to a coarse-navigation location at a locale of a cluster of charging pads, the cluster comprising the charging pads arranged in a layout, and a plurality of fiducial markers distributed at positions across the layout; cause the imaging system to acquire an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads, wherein the acquired image captures an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location; in a reference map of the layout and a reference sub-plurality of the fiducial markers, identify at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map; and based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence, disambiguate an identity of a particular charging pad from among the one or more charging pads.

In a second aspect, a system may comprise an uncrewed aerial vehicle (UAV), a server in an infrastructure support network for UAVs, and a cluster of charging pads for UAVs. The UAV may comprise a navigation system, an imaging system, and a control system, and the cluster may comprise the charging pads arranged in a layout and a plurality of fiducial markers distributed at positions across the layout. The control system configured of the UAV may be configured to: cause the UAV to fly to a locale of the cluster of charging pads; cause the imaging system to acquire at least one aerial image of the cluster, wherein the acquired at least one aerial image captures an observed constellation of fiducial markers at apparent positions and orientations relative to one or more of the charging pads, as viewed from at least one position and orientation of the UAV with respect to the cluster; and send data comprising the at least one aerial image to the server. The server may be configured to: receive the at least one aerial image; generate and/or update a reference map of the cluster based on the at least one aerial image; and provide the reference map to UAVs having missions that include flying to the cluster of charging pads.

In a third aspect, a method carried out by an uncrewed aerial vehicle (UAV) may comprise: receiving a reference map of a cluster of charging pads from a server in an infrastructure support network for UAVs, the cluster comprising the charging pads arranged in a layout, and a plurality of fiducial markers distributed at positions across the layout, and the reference map representing the layout and a reference sub-plurality of the fiducial markers; storing the reference map in a memory of the UAV; flying to a coarse-navigation location at a locale of the cluster of charging pads; with an imaging system of the UAV, acquiring an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads, wherein the acquired image captures an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location; in the reference map, identifying at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map; and based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence, disambiguating an identity of a particular charging pad from among the one or more charging pads.

In a fourth aspect, a non-transitory computer-readable medium may store instructions that, when executed by one or more processors of a control system of an uncrewed aerial vehicle (UAV), may cause the control system to carry out operations including: receiving a reference map of a cluster of charging pads from a server in an infrastructure support network for UAVs, the cluster comprising the charging pads arranged in a layout, and a plurality of fiducial markers distributed at positions across the layout, and the reference map representing the layout and a reference sub-plurality of the fiducial markers; storing the reference map in a memory of the UAV; flying to a coarse-navigation location at a locale of the cluster of charging pads; with an imaging system of the UAV, acquiring an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads, wherein the acquired image captures an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location; in the reference map, identifying at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map; and based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence, disambiguating an identity of a particular charging pad from among the one or more charging pads.

In a fifth aspect, a system comprising an uncrewed aerial vehicle (UAV), a server in an infrastructure support network for UAVs, and a cluster of charging pads for UAVs is provided. The UAV may comprise a navigation system, an imaging system, and a control system, and the cluster may comprise the charging pads arranged in a layout and a plurality of fiducial markers distributed at positions across the layout. The system may include means for carrying out various operations including: causing the UAV to fly to a locale of the cluster of charging pads; causing the imaging system to acquire at least one aerial image of the cluster, wherein the acquired at least one aerial image captures an observed constellation of fiducial markers at apparent positions and orientations relative to one or more of the charging pads, as viewed from at least one position and orientation of the UAV with respect to the cluster; and causing the UAV to send data comprising the at least one aerial image to the server. The operations may further include: causing the server to receive the at least one aerial image; causing the server to generate and/or update a reference map of the cluster based on the at least one aerial image; and causing the server to provide the reference map to UAVs having missions that include flying to the cluster of charging pads.

In a sixth aspect, a system is provided that includes means for causing an uncrewed aerial vehicle (UAV) to carry out operations including: receiving a reference map of a cluster of charging pads from a server in an infrastructure support network for UAVs, the cluster comprising the charging pads arranged in a layout, and a plurality of fiducial markers distributed at positions across the layout, and the reference map representing the layout and a reference sub-plurality of the fiducial markers; storing the reference map in a memory of the UAV; flying to a coarse-navigation location at a locale of the cluster of charging pads; with an imaging system of the UAV, acquiring an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads, wherein the acquired image captures an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location; in the reference map, identifying at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map; and based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence, disambiguating an identity of a particular charging pad from among the one or more charging pads.

These, as well as other aspects, advantages, and alternatives, will become apparent to those of ordinary skill in the art by reading the following detailed description with reference where appropriate to the accompanying drawings. Further, it should be understood that the description provided in this summary section and elsewhere in this document is intended to illustrate the claimed subject matter by way of example and not by way of limitation.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A is a simplified illustration of an uncrewed aerial vehicle, according to example embodiments, in accordance with example embodiments.

FIG. **1**B is a simplified illustration of an uncrewed aerial vehicle, according to example embodiments, in accordance with example embodiments.

FIG. **1**C is a simplified illustration of an uncrewed aerial vehicle, according to example embodiments, in accordance with example embodiments.

FIG. **1**D is a simplified illustration of an uncrewed aerial vehicle, according to example embodiments, in accordance with example embodiments.

FIG. **1**E is a simplified illustration of an uncrewed aerial vehicle, according to example embodiments, in accordance with example embodiments.

FIG. **2** is a simplified block diagram illustrating components of an uncrewed aerial vehicle, in accordance with example embodiments.

FIG. **3** is a simplified block diagram illustrating a UAV system, in accordance with example embodiments.

FIG. **4**A illustrates an example charging pad cluster, in accordance with example embodiments.

FIG. **4**B is a representation of nominal UAV navigational accuracy in relation to an example charging pad cluster, in accordance with example embodiments.

FIG. **4**C illustrates an example charging pad cluster with example fiducial markers, in accordance with example embodiments.

FIG. **4**D illustrates an example charging pad cluster with example fiducial markers as viewed from different hypothetical perspectives, in accordance with example embodiments.

FIG. **5** is representation of an example reference map of an example charging pad cluster and example reference constellations of fiducial markers, in accordance with example embodiments.

FIG. **6** is a conceptual illustration of an example operation for disambiguating charging pad identities, in accordance with example embodiments.

FIG. **7** is a block diagram of a method, in accordance with example embodiments.

FIG. **8** is a block diagram of another method, in accordance with example embodiments.

DETAILED DESCRIPTION

Exemplary methods and systems are described herein. It should be understood that the word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any implementation or feature described herein as "exemplary" or "illustrative" is not necessarily to be construed as preferred or advantageous over other implementations or features. In the figures, similar symbols typically identify similar components, unless context dictates otherwise. The example implementations described herein are not meant to be limiting. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are contemplated herein.

I. OVERVIEW

An example usage of UA Vs may be to deliver various items to customers. For example, UAVs may be tasked with picking up a payload containing an item from a location and delivering the payload to a customer's residence, commercial building, or other location. In a typical usage scenario a group or fleet of UAVs may be deployed by a fleet operator to provide a delivery service for its own product offerings and/or for other vendors. A fleet of UAVs may be used for additional and/or other services, such as environment surveillance and emergency services support, among other non-limiting examples. The tasks associated with services provided by a fleet of UA Vs may sometimes be referred to herein as "missions," a term meant to convey that a set of operational steps that may be involved in carrying out a task, such as delivery.

In order to support a fleet of UAVs in their missions, as well as support the service or services provided by the fleet, a fleet operator may also deploy an infrastructure. As described in detail below, such an "infrastructure support network," as it is sometimes referred to herein, may include servers and computing systems for managing mission operations, coordinating UAV tasks, interfacing with customers (or other service recipients), and maintaining fleet health and readiness. In accordance with example embodiments, the infrastructure may be implemented in a centralized and/or distributed fashion, and may be organized hierarchically. Infrastructure may also include and or be connected with one or more communications networks, such as a public and/or private internet, for supporting communications between other infrastructure elements.

One aspect of maintaining fleet health and readiness may entail charging batteries of individual UAVs. In accordance with example embodiments, battery charging may be implemented using large-scale deployments of charging pads arranged in groups or clusters located at various sites across an operating region, such as a city or metropolitan area. For example, each vendor using a UAV delivery service may host a cluster of 10-50 or more charging pads at each of its vendor sites. Each cluster may be powered by available AC (alternating current) power, and be relatively inexpensive to operate, incurring only electric utility fees for intermittent usage, for example. Charging pads may also serve as origination points and/or termination points of UAV missions and/or mission legs (or segments), such as delivery pick-up and/or drop-off points.

In accordance with example embodiments, charging pad clusters may be deployed as standalone sites and/or integrated with additional infrastructure elements or components for supporting other aspects of UAV fleet operation. For example, an aspect of mission operations may involve collecting and processing/analyzing various forms of operations reporting data from UAVs accumulated during missions, and providing various forms of operations configuration data to UAVs. Thus, infrastructure facilities for bulk data transfers between UAV and fleet support systems may include a wireless access point communicatively connected to a wireless and/or wireline broadband interface to a data backhaul network, such as a data/telecommunications carrier core network and/or internet. The data backhaul network may then be communicatively connected to servers and computer systems of the fleet support infrastructure. A UAV may communicate via a wireless connection with the wireless access point. In some instances, a UAV may have its own cellular wireless interface to a cellular data and telecommunications network.

Such "full-service" operations support facilities are sometimes referred to as "jet bridges," in analogy to like-named facilities where commercial aircraft typically park at airport terminals. Thus, a jet bridge may provide one or more charging pads, as well as a communicative connection to an infrastructure support network. An infrastructure support network may include one or more servers that support flight scheduling, flight plans, and mission plans, for example. Jet bridges and/or standalone charging pad clusters may be deployed a warehouses, retail shopping centers, and/or other facilities that may be served by UAVs and/or support UAV operations.

A UAV may have a navigation system that provides a relatively coarse mode of navigational accuracy to enable the UAV to achieve, with a specified likelihood of navigational success, flying to within a threshold tolerance of a designated target location. The coarse mode of navigation may use GPS, for example, which could have limited and/or impeded accuracy in some circumstances. In an example usage scenario, the designated target location may be a designated target charging pad within a cluster of charging pads arranged in a grid or other layout, and the specified likelihood and threshold tolerance may be such that the UAV may have a non-vanishing likelihood of arriving at the wrong charging pad. In the context of a grid (or other layout) of charging pads, the UAV may thus need to augment its coarse navigational solution in order to be able to distinguish the correct, designated target charging pad from among other nearby pads that may be within the threshold tolerance. Example embodiments herein provide the augmented (or additional) precision navigation that provides a UAV with the capability to determine the requisite disambiguation.

Accordingly, a UAV may have a mission that includes flying to and (possibly) landing on a designated charging pad in a cluster of charging pads having a distribution of fiducial markers at various positions and orientations relative to the charging pads. In accordance with example embodiments, the UAV may be provided with a reference map of the cluster, including the fiducial markers. The reference map may represent a view of the cluster from above in which the spatial distribution of fiducial markers in the vicinity of any given charging pad form a sort of reference constellation or pattern that is distinguishable from similarly-formed reference constellations in the vicinities of other charging pads of the cluster. The UAV may thus use its coarse navigation mode to fly to and arrive at a coarse-navigation position within the threshold tolerance of the designated target charging pad, and then use an imaging system to acquire an image of the cluster from its perspective at the coarse-navigation position and orientation. The UAV may identify one or more observed charging pads in the image, as well as corresponding observed constellations of fiducial markers for each of the one or more observed charging pads. By applying one or another form a pattern matching operation to one or more of the observed constellations and one or more of the reference constellations in the reference map, the UAV may then distinguish or disambiguate between the identities or cluster positions of the one or more observed charging pads. In particular, the UAV may use the disambiguation to identify the designated target charging pad and/or disqualify a different observed charging pad that might otherwise be misidentified as the designated target charging pad.

In some deployments, the charging pads of a cluster may be arranged in a layout, such as a regular grid or array. In accordance with example embodiments, by using constellations of fiducial markers at their respective positions and orientations with respect to charging pads of a cluster, the fiducial markers need not necessarily be deployed at precise geolocations and/or have precisely determined geolocations. In some examples, the distribution and orientations of fiducial markers at locations across a cluster may be random and/or regular. Additionally, a match between a reference constellation and an observed constellation need not necessarily be exact in order to confirm or disqualify an identification. Thus, this technique may tolerate some degree of discrepancy between a reference map of a cluster and the actual distribution of fiducial markers observed or captured in the UAV's image. Discrepancies could result from damage to, and/or debris on, one or more fiducial markers, for example. Some discrepancies may result, at least in part, simply from misalignment between observed and reference constellations. Straightforward geometric transformations (e.g., rotations and/or translations) may be applied to compensate for such misalignments (or other differences of perspective between observed and reference constellations).

Further, to the extent that the fiducial markers may be identified by visibly (or otherwise) recognizable physical features, all of the fiducial markers of a cluster need not necessarily have unique identifications in order for different constellations to be distinguishable. Thus, by using recognizable constellations of fiducial markers to disambiguate among charging pads, charging pad clusters including fiducial markers may be deployed without necessarily requiring determination of precise geolocations of pads or fiducial markers, or necessarily requiring unique identifications among fiducial markers of the cluster. As resulting benefit, it may also be possible to reduce the need for on-site maintenance of fiducial markers compared to what might otherwise be required.

Because constellations or patterns of fiducial markers with respect to charging pads of a cluster derive from relative positions and orientations of the fiducial markers with respect to the charging pads, and do not necessarily depend on precise geolocations of the fiducial markers, the term "context-based navigation" may be used to describe the techniques and methodology of example embodiments herein. That is, recognizable patterns of fiducial markers may provide a context for positional and orientational awareness of a UAV with respect to specific charging pads of a cluster that does not necessarily require a determination of the UAV's precise geolocation.

In further accordance with example embodiments, reference maps of a cluster may be generated by a server or other processing system from aerial surveillance images obtained from planned or incidental flights of UAVs over the cluster. For example, one or more UAVs may have missions that include acquiring aerial images of one or more clusters from various positions and perspectives above the clusters. The images may be sent—e.g., in data transmissions—to a server in an infrastructure support network for UAVs, which may then synthesize (or otherwise process) an optimized aerial view of the cluster, and generate a reference map from the optimized view. A reference map may be updated and/or revised from time to time by additional aerial images acquired by other UAVs having missions that include flying to the cluster. Updating reference maps in this way could also help account for intended and/or unintended changes to the distribution of fiducial markers at a cluster. Further details and advantages of example embodiments are described below.

II. EXAMPLE UNCREWED VEHICLES

Herein, the terms "uncrewed aerial vehicle" and "UAV" refer to any autonomous or semi-autonomous vehicle that is capable of performing some functions without a physically present human pilot. As noted above, the term "unmanned" may sometimes be used instead of, or in addition to, "uncrewed," and it should be understood that both terms have the same meaning, and may be used interchangeably.

A UAV can take various forms. For example, a UAV may take the form of a fixed-wing aircraft, a glider aircraft, a tail-sitter aircraft, a jet aircraft, a ducted fan aircraft, a lighter-than-air dirigible such as a blimp or steerable balloon, a rotorcraft such as a helicopter or multicopter, and/or an ornithopter, among other possibilities. Further, the terms "drone," "uncrewed aerial vehicle system" (UAVS), or "uncrewed aerial system" (UAS) may also be used to refer to a UAV.

FIG. **1**A is an isometric view of an example UAV **100**. UAV **100** includes wing **102**, booms **104**, and a fuselage **106**. Wings **102** may be non-flying stationary and may generate lift based on the wing shape and the UAV's forward airspeed. For instance, the two wings **102** may have an airfoil-shaped cross section to produce an aerodynamic force on UAV **100**. In some embodiments, wing **102** may carry horizontal propulsion units **108**, and booms **104** may carry vertical propulsion units **110**. In operation, power for the propulsion units may be provided from a battery compartment **112** of fuselage **106**. In some embodiments, fuselage **106** also includes an avionics compartment **114**, an additional battery compartment (not shown) and/or a delivery unit (not shown, e.g., a winch system) for handling the payload. In some embodiments, fuselage **106** is modular, and two or more compartments (e.g., battery compartment **112**, avionics compartment **114**, other payload and delivery compartments) are detachable from each other and securable to each other (e.g., mechanically, magnetically, or otherwise) to contiguously form at least a portion of fuselage **106**.

In some embodiments, booms **104** terminate in rudders **116** for improved yaw control of UAV **100**. Further, wings **102** may terminate in wing tips **117** for improved control of lift of the UAV.

In the illustrated configuration, UAV **100** includes a structural frame. The structural frame may be referred to as a "structural H-frame" or an "H-frame" (not shown) of the UAV. The H-frame may include, within wings **102**, a wing spar (not shown) and, within booms **104**, boom carriers (not shown). In some embodiments the wing spar and the boom carriers may be made of carbon fiber, hard plastic, aluminum, light metal alloys, or other materials. The wing spar and the boom carriers may be connected with clamps. The wing spar may include pre-drilled holes for horizontal propulsion units **108**, and the boom carriers may include pre-drilled holes for vertical propulsion units **110**.

In some embodiments, fuselage **106** may be removably attached to the H-frame (e.g., attached to the wing spar by clamps, configured with grooves, protrusions or other features to mate with corresponding H-frame features, etc.). In other embodiments, fuselage **106** similarly may be removably attached to wings **102**. The removable attachment of fuselage **106** may improve quality and or modularity of UAV **100**. For example, electrical/mechanical components and/or subsystems of fuselage **106** may be tested separately from, and before being attached to, the H-frame. Similarly, printed circuit boards (PCBs) **118** may be tested separately from, and before being attached to, the boom carriers, therefore eliminating defective parts/subassemblies prior to completing the UAV. For example, components of fuselage **106** (e.g., avionics, battery unit, delivery units, an additional battery compartment, etc.) may be electrically tested before fuselage **106** is mounted to the H-frame. Furthermore, the motors and the electronics of PCBs **118** may also be electrically tested before the final assembly. Generally, the identification of the defective parts and subassemblies early in the assembly process lowers the overall cost and lead time of the UAV. Furthermore, different types/models of fuselage **106** may be attached to the H-frame, therefore improving the modularity of the design. Such modularity allows these various parts of UAV **100** to be upgraded without a substantial overhaul to the manufacturing process.

In some embodiments, a wing shell and boom shells may be attached to the H-frame by adhesive elements (e.g., adhesive tape, double-sided adhesive tape, glue, etc.). Therefore, multiple shells may be attached to the H-frame instead of having a monolithic body sprayed onto the H-frame. In some embodiments, the presence of the multiple shells reduces the stresses induced by the coefficient of thermal expansion of the structural frame of the UAV. As a result, the UAV may have better dimensional accuracy and/or improved reliability.

Moreover, in at least some embodiments, the same H-frame may be used with the wing shell and/or boom shells having different size and/or design, therefore improving the modularity and versatility of the UAV designs. The wing shell and/or the boom shells may be made of relatively light polymers (e.g., closed cell foam) covered by the harder, but relatively thin, plastic skins.

The power and/or control signals from fuselage **106** may be routed to PCBs **118** through cables running through fuselage **106**, wings **102**, and booms **104**. In the illustrated embodiment, UAV **100** has four PCBs, but other numbers of PCBs are also possible. For example, UAV **100** may include two PCBs, one per the boom. The PCBs carry electronic components **119** including, for example, power converters, controllers, memory, passive components, etc. In operation, propulsion units **108** and **110** of UAV **100** are electrically connected to the PCBs.

Many variations on the illustrated UAV are possible. For instance, fixed-wing UAVs may include more or fewer rotor units (vertical or horizontal), and/or may utilize a ducted fan or multiple ducted fans for propulsion. Further, UAVs with more wings (e.g., an "x-wing" configuration with four wings), are also possible. Although FIG. **1** illustrates two wings **102**, two booms **104**, two horizontal propulsion units **108**, and six vertical propulsion units **110** per boom **104**, it should be appreciated that other variants of UAV **100** may be implemented with more or less of these components. For example, UAV **100** may include four wings **102**, four booms **104**, and more or less propulsion units (horizontal or vertical).

Similarly, FIG. **1**B shows another example of a fixed-wing UAV **120**. The fixed-wing UAV **120** includes a fuselage **122**, two wings **124** with an airfoil-shaped cross section to provide lift for the UAV **120**, a vertical stabilizer **126** (or fin) to stabilize the plane's yaw (turn left or right), a horizontal stabilizer **128** (also referred to as an elevator or tailplane) to stabilize pitch (tilt up or down), landing gear **130**, and a propulsion unit **132**, which can include a motor, shaft, and propeller.

FIG. **1**C shows an example of a UAV **140** with a propeller in a pusher configuration. The term "pusher" refers to the fact that a propulsion unit **142** is mounted at the back of the UAV and "pushes" the vehicle forward, in contrast to the propulsion unit being mounted at the front of the UAV. Similar to the description provided for FIGS. **1**A and **1**B, FIG. **1**C depicts common structures used in a pusher plane, including a fuselage **144**, two wings **146**, vertical stabilizers **148**, and the propulsion unit **142**, which can include a motor, shaft, and propeller.

FIG. **1**D shows an example of a tail-sitter UAV **160**. In the illustrated example, the tail-sitter UAV **160** has fixed wings **162** to provide lift and allow the UAV **160** to glide horizontally (e.g., along the x-axis, in a position that is approximately perpendicular to the position shown in FIG. **1**D). However, the fixed wings **162** also allow the tail-sitter UAV **160** to take off and land vertically on its own.

For example, at a launch site, the tail-sitter UAV **160** may be positioned vertically (as shown) with its fins **164** and/or wings **162** resting on the ground and stabilizing the UAV **160** in the vertical position. The tail-sitter UAV **160** may then take off by operating its propellers **166** to generate an upward thrust (e.g., a thrust that is generally along the y-axis). Once at a suitable altitude, the tail-sitter UAV **160** may use its flaps **168** to reorient itself in a horizontal position, such that its fuselage **170** is closer to being aligned with the x-axis than the y-axis. Positioned horizontally, the propellers **166** may provide forward thrust so that the tail-sitter UAV **160** can fly in a similar manner as a typical airplane.

Many variations on the illustrated fixed-wing UAVs are possible. For instance, fixed-wing UAVs may include more or fewer propellers, and/or may utilize a ducted fan or multiple ducted fans for propulsion. Further, UAVs with more wings (e.g., an "x-wing" configuration with four wings), with fewer wings, or even with no wings, are also possible.

As noted above, some embodiments may involve other types of UAVs, in addition to or in the alternative to fixed-wing UAVs. For instance, FIG. **1**E shows an example of a rotorcraft that is commonly referred to as a multicopter **180**. The multicopter **180** may also be referred to as a quadcopter, as it includes four rotors **182**. It should be understood that example embodiments may involve a rotorcraft with more or fewer rotors than the multicopter **180**. For example, a helicopter typically has two rotors. Other examples with three or more rotors are possible as well. Herein, the term "multicopter" refers to any rotorcraft having more than two rotors, and the term "helicopter" refers to rotorcraft having two rotors.

Referring to the multicopter **180** in greater detail, the four rotors **182** provide propulsion and maneuverability for the multicopter **180**. More specifically, each rotor **182** includes blades that are attached to a motor **184**. Configured as such, the rotors **182** may allow the multicopter **180** to take off and land vertically, to maneuver in any direction, and/or to hover. Further, the pitch of the blades may be adjusted as a group and/or differentially, and may allow the multicopter **180** to control its pitch, roll, yaw, and/or altitude.

It should be understood that references herein to an "uncrewed" aerial vehicle or UAV can apply equally to autonomous and semi-autonomous aerial vehicles. In an autonomous implementation, all functionality of the aerial vehicle is automated; e.g., pre-programmed or controlled via real-time computer functionality that responds to input from various sensors and/or pre-determined information. In a semi-autonomous implementation, some functions of an aerial vehicle may be controlled by a human operator, while other functions are carried out autonomously. Further, in some embodiments, a UAV may be configured to allow a remote operator to take over functions that can otherwise be controlled autonomously by the UAV. Yet further, a given type of function may be controlled remotely at one level of abstraction and performed autonomously at another level of abstraction. For example, a remote operator could control high level navigation decisions for a UAV, such as by specifying that the UAV should travel from one location to another (e.g., from a warehouse in a suburban area to a delivery address in a nearby city), while the UAV's navigation system autonomously controls more fine-grained navigation decisions, such as the specific route to take between the two locations, specific flight controls to achieve the route and avoid obstacles while navigating the route, and so on.

More generally, it should be understood that the example UAVs described herein are not intended to be limiting. Example embodiments may relate to, be implemented within, or take the form of any type of uncrewed aerial vehicle.

III. ILLUSTRATIVE UAV COMPONENTS

FIG. **2** is a simplified block diagram illustrating components of a UAV **200**, according to an example embodiment. UAV **200** may take the form of, or be similar in form to, one of the UAVs **100**, **120**, **140**, **160**, and **180** described in reference to FIGS. **1A-1E**. However, UAV **200** may also take other forms.

UAV **200** may include various types of sensors, and may include a computing system configured to provide the functionality described herein. In the illustrated embodiment, the sensors of UAV **200** include an inertial measurement unit (IMU) **202**, ultrasonic sensor(s) **204**, and a GPS **206**, among other possible sensors and sensing systems.

In the illustrated embodiment, UAV **200** also includes one or more processors **208**. A processor **208** may be a general-purpose processor or a special purpose processor (e.g., digital signal processors, application specific integrated circuits, etc.). The one or more processors **208** can be configured to execute computer-readable program instructions **212** that are stored in the data storage **210** and are executable to provide the functionality of a UAV described herein.

The data storage **210** may include or take the form of one or more computer-readable storage media that can be read or accessed by at least one processor **208**. The one or more computer-readable storage media can include volatile and/or non-volatile storage components, such as optical, magnetic, organic or other memory or disc storage, which can be integrated in whole or in part with at least one of the one or more processors **208**. In some embodiments, the data storage **210** can be implemented using a single physical device (e.g., one optical, magnetic, organic or other memory or disc storage unit), while in other embodiments, the data storage **210** can be implemented using two or more physical devices.

As noted, the data storage **210** can include computer-readable program instructions **212** and perhaps additional data, such as diagnostic data of the UAV **200**. As such, the data storage **210** may include program instructions **212** to perform or facilitate some or all of the UAV functionality described herein. For instance, in the illustrated embodiment, program instructions **212** include a navigation module **214** and a tether control module **216**.

A. Sensors

In an illustrative embodiment, IMU **202** may include both an accelerometer and a gyroscope, which may be used together to determine an orientation of the UAV **200**. In particular, the accelerometer can measure the orientation of the vehicle with respect to earth, while the gyroscope measures the rate of rotation around an axis. IMUs are commercially available in low-cost, low-power packages. For instance, an IMU **202** may take the form of or include a miniaturized MicroElectroMechanical System (MEMS) or a NanoElectroMechanical System (NEMS). Other types of IMUs may also be utilized.

An IMU **202** may include other sensors, in addition to accelerometers and gyroscopes, which may help to better determine position and/or help to increase autonomy of the UAV **200**. Two examples of such sensors are magnetometers and pressure sensors. In some embodiments, a UAV may include a low-power, digital 3-axis magnetometer, which can be used to realize an orientation independent electronic compass for accurate heading information. However, other types of magnetometers may be utilized as well. Other examples are also possible. Further, note that a UAV could include some or all of the above-described inertia sensors as separate components from an IMU.

UAV **200** may also include a pressure sensor or barometer, which can be used to determine the altitude of the UAV **200**. Alternatively, other sensors, such as sonic altimeters or radar altimeters, can be used to provide an indication of altitude, which may help to improve the accuracy of and/or prevent drift of an IMU.

In a further aspect, UAV **200** may include one or more sensors that allow the UAV to sense objects in the environment. For instance, in the illustrated embodiment, UAV **200** includes ultrasonic sensor(s) **204**. Ultrasonic sensor(s) **204** can determine the distance to an object by generating sound waves and determining the time interval between transmission of the wave and receiving the corresponding echo off an object. A typical application of an ultrasonic sensor for uncrewed vehicles or IMUs is low-level altitude control and obstacle avoidance. An ultrasonic sensor can also be used for vehicles that need to hover at a certain height or need to be capable of detecting obstacles. Other systems can be used to determine, sense the presence of, and/or determine the distance to nearby objects, such as a light detection and ranging (LIDAR) system, laser detection and ranging (LADAR) system, and/or an infrared or forward-looking infrared (FLIR) system, among other possibilities.

In some embodiments, UAV **200** may also include one or more imaging system(s). For example, one or more still and/or video cameras may be utilized by UAV **200** to capture image data from the UAV's environment. As a specific example, charge-coupled device (CCD) cameras or complementary metal-oxide-semiconductor (CMOS) cameras can be used with uncrewed vehicles. Such imaging sensor(s) have numerous possible applications, such as obstacle avoidance, localization techniques, ground tracking for more accurate navigation (e.g., by applying optical flow techniques to images), video feedback, and/or image recognition and processing, among other possibilities.

UAV **200** may also include a GPS receiver **206**. The GPS receiver **206** may be configured to provide data that is typical of well-known GPS systems, such as the GPS coordinates of the UAV **200**. Such GPS data may be utilized by the UAV **200** for various functions. As such, the UAV may use its GPS receiver **206** to help navigate to the caller's location, as indicated, at least in part, by the GPS coordinates provided by their mobile device. Other examples are also possible.

B. Navigation and Location Determination

The navigation module **214** may provide functionality that allows the UAV **200** to, e.g., move about its environment and reach a desired location. To do so, the navigation module **214** may control the altitude and/or direction of flight by controlling the mechanical features of the UAV that affect flight (e.g., its rudder(s), elevator(s), aileron(s), and/or the speed of its propeller(s)).

In order to navigate the UAV **200** to a target location, the navigation module **214** may implement various navigation techniques, such as map-based navigation and localization-based navigation, for instance. With map-based navigation, the UAV **200** may be provided with a map of its environment, which may then be used to navigate to a particular location on the map. With localization-based navigation, the UAV **200** may be capable of navigating in an unknown environment using localization. Localization-based navigation may involve the UAV **200** building its own map of its environment and calculating its position within the map and/or the position of objects in the environment. For example, as a UAV **200** moves throughout its environment, the UAV **200** may continuously use localization to update its map of the environment. This continuous mapping process may be referred to as simultaneous localization and mapping (SLAM). Other navigation techniques may also be utilized.

In some embodiments, the navigation module **214** may navigate using a technique that relies on waypoints. In particular, waypoints are sets of coordinates that identify points in physical space. For instance, an air-navigation waypoint may be defined by a certain latitude, longitude, and altitude. Accordingly, navigation module **214** may cause UAV **200** to move from waypoint to waypoint, in order to ultimately travel to a final destination (e.g., a final waypoint in a sequence of waypoints).

In a further aspect, the navigation module **214** and/or other components and systems of the UAV **200** may be configured for "localization" to more precisely navigate to the scene of a target location. More specifically, it may be desirable in certain situations for a UAV to be within a threshold distance of the target location where a payload **228** is being delivered by a UAV (e.g., within a few feet of the target destination). To this end, a UAV may use a two-tiered approach in which it uses a more-general location-determination technique to navigate to a general area that is associated with the target location, and then use a more-refined location-determination technique to identify and/or navigate to the target location within the general area.

For example, the UAV **200** may navigate to the general area of a target destination where a payload **228** is being delivered using waypoints and/or map-based navigation. The UAV may then switch to a mode in which it utilizes a localization process to locate and travel to a more specific location. For instance, if the UAV **200** is to deliver a payload to a user's home, the UAV **200** may need to be substantially close to the target location in order to avoid delivery of the payload to undesired areas (e.g., onto a roof, into a pool, onto a neighbor's property, etc.). However, a GPS signal may only get the UAV **200** so far (e.g., within a block of the user's home). A more precise location-determination technique may then be used to find the specific target location.

Various types of location-determination techniques may be used to accomplish localization of the target delivery location once the UAV **200** has navigated to the general area of the target delivery location. For instance, the UAV **200** may be equipped with one or more sensory systems, such as, for example, ultrasonic sensors **204**, infrared sensors (not shown), and/or other sensors, which may provide input that the navigation module **214** utilizes to navigate autonomously or semi-autonomously to the specific target location.

As another example, once the UAV **200** reaches the general area of the target delivery location (or of a moving subject such as a person or their mobile device), the UAV **200** may switch to a "fly-by-wire" mode where it is controlled, at least in part, by a remote operator, who can navigate the UAV **200** to the specific target location. To this end, sensory data from the UAV **200** may be sent to the remote operator to assist them in navigating the UAV **200** to the specific location.

As yet another example, the UAV **200** may include a module that is able to signal to a passer-by for assistance in either reaching the specific target delivery location; for example, the UAV **200** may display a visual message requesting such assistance in a graphic display, play an audio message or tone through speakers to indicate the need for such assistance, among other possibilities. Such a visual or audio message might indicate that assistance is needed in delivering the UAV **200** to a particular person or a particular location, and might provide information to assist the passer-by in delivering the UAV **200** to the person or location (e.g., a description or picture of the person or location, and/or the person or location's name), among other possibilities. Such a feature can be useful in a scenario in which the UAV is unable to use sensory functions or another location-determination technique to reach the specific target location. However, this feature is not limited to such scenarios.

In some embodiments, once the UAV **200** arrives at the general area of a target delivery location, the UAV **200** may utilize a beacon from a user's remote device (e.g., the user's mobile phone) to locate the person. Such a beacon may take various forms. As an example, consider the scenario where a remote device, such as the mobile phone of a person who requested a UAV delivery, is able to send out directional signals (e.g., via an RF signal, a light signal and/or an audio signal). In this scenario, the UAV **200** may be configured to navigate by "sourcing" such directional signals—in other words, by determining where the signal is strongest and navigating accordingly. As another example, a mobile device can emit a frequency, either in the human range or outside the human range, and the UAV **200** can listen for that frequency and navigate accordingly. As a related example, if the UAV **200** is listening for spoken commands, then the UAV **200** could utilize spoken statements, such as "I'm over here!" to source the specific location of the person requesting delivery of a payload.

In an alternative arrangement, a navigation module may be implemented at a remote computing device, which communicates wirelessly with the UAV **200**. The remote computing device may receive data indicating the operational state of the UAV **200**, sensor data from the UAV **200** that allows it to assess the environmental conditions being experienced by the UAV **200**, and/or location information for the UAV **200**. Provided with such information, the remote computing device may determine latitudinal and/or directional adjustments that should be made by the UAV **200** and/or may determine how the UAV **200** should adjust its mechanical features (e.g., its rudder(s), elevator(s), aileron(s), and/or the speed of its propeller(s)) in order to effectuate such movements. The remote computing system may then communicate such adjustments to the UAV **200** so it can move in the determined manner.

C. Communication Systems

In a further aspect, the UAV **200** includes one or more communication systems **218**. The communications systems **218** may include one or more wireless interfaces and/or one or more wireline interfaces, which allow the UAV **200** to communicate via one or more networks. Such wireless interfaces may provide for communication under one or more wireless communication protocols, such as Bluetooth, WiFi (e.g., an IEEE 802.11 protocol), Long-Term Evolution (LTE), WiMAX (e.g., an IEEE 802.16 standard), a radio-frequency ID (RFID) protocol, near-field communication (NFC), and/or other wireless communication protocols. Such wireline interfaces may include an Ethernet interface, a Universal Serial Bus (USB) interface, or similar interface to communicate via a wire, a twisted pair of wires, a coaxial cable, an optical link, a fiber-optic link, or other physical connection to a wireline network.

In some embodiments, a UAV **200** may include communication systems **218** that allow for both short-range communication and long-range communication. For example, the UAV **200** may be configured for short-range communications using Bluetooth and for long-range communications under a CDMA protocol. In such an embodiment, the UAV **200** may be configured to function as a "hot spot;" or in other words, as a gateway or proxy between a remote support device and one or more data networks, such as a cellular network and/or the Internet. Configured as such, the UAV **200** may facilitate data communications that the remote support device would otherwise be unable to perform by itself.

For example, the UAV **200** may provide a WiFi connection to a remote device, and serve as a proxy or gateway to a cellular service provider's data network, which the UAV might connect to under an LTE or a 3G protocol, for instance. The UAV **200** could also serve as a proxy or gateway to a high-altitude balloon network, a satellite network, or a combination of these networks, among others, which a remote device might not be able to otherwise access.

D. Power Systems

In a further aspect, the UAV **200** may include power system(s) **220**. The power system **220** may include one or more batteries for providing power to the UAV **200**. In one example, the one or more batteries may be rechargeable and each battery may be recharged via a wired connection between the battery and a power supply and/or via a wireless charging system, such as an inductive charging system that applies an external time-varying magnetic field to an internal battery.

E. Payload Delivery

The UAV **200** may employ various systems and configurations in order to transport and deliver a payload **228**. In some implementations, the payload **228** of a given UAV **200** may include or take the form of a "package" designed to transport various goods to a target delivery location. For example, the UAV **200** can include a compartment, in which an item or items may be transported. Such a package may include one or more food items, purchased goods, medical items, or any other object(s) having a size and weight suitable to be transported between two locations by the UAV. In other embodiments, a payload **228** may simply be the one or more items that are being delivered (e.g., without any package housing the items).

In some embodiments, the payload **228** may be attached to the UAV and located substantially outside of the UAV during some or all of a flight by the UAV. For example, the package may be tethered or otherwise releasably attached below the UAV during flight to a target location. In some embodiments, the package may include various features that protect its contents from the environment, reduce aerodynamic drag on the system, and prevent the contents of the package from shifting during UAV flight. In other embodiments, the package may be a standard shipping package that is not specifically tailored for UAV flight.

In order to deliver the payload, the UAV may include a winch system **221** controlled by the tether control module **216** in order to lower the payload **228** to the ground while the UAV hovers above. As shown in FIG. **2**, the winch system **221** may include a tether **224**, and the tether **224** may be coupled to the payload **228** by a payload retriever **226**. The tether **224** may be wound on a spool that is coupled to a motor **222** of the UAV. The motor **222** may take the form of a DC motor (e.g., a servo motor) that can be actively controlled by a speed controller. The tether control module **216** can control the speed controller to cause the motor **222** to rotate the spool, thereby unwinding or retracting the tether **224** and lowering or raising the payload retriever **226**. In practice, the speed controller may output a desired operating rate (e.g., a desired RPM) for the spool, which may correspond to the speed at which the tether **224** and payload **228** should be lowered towards the ground. The motor **222** may then rotate the spool so that it maintains the desired operating rate.

In order to control the motor **222** via the speed controller, the tether control module **216** may receive data from a speed sensor (e.g., an encoder) configured to convert a mechanical position to a representative analog or digital signal. In particular, the speed sensor may include a rotary encoder that may provide information related to rotary position (and/or rotary movement) of a shaft of the motor or the spool coupled to the motor, among other possibilities. Moreover, the speed sensor may take the form of an absolute encoder and/or an incremental encoder, among others. So in an example implementation, as the motor **222** causes rotation of the spool, a rotary encoder may be used to measure this rotation. In doing so, the rotary encoder may be used to convert a rotary position to an analog or digital electronic signal used by the tether control module **216** to determine the amount of rotation of the spool from a fixed reference angle and/or to an analog or digital electronic signal that is representative of a new rotary position, among other options. Other examples are also possible.

Based on the data from the speed sensor, the tether control module **216** may determine a rotational speed of the motor **222** and/or the spool and responsively control the motor **222** (e.g., by increasing or decreasing an electrical current supplied to the motor **222**) to cause the rotational speed of the motor **222** to match a desired speed. When adjusting the motor current, the magnitude of the current adjustment may be based on a proportional-integral-derivative (PID) calculation using the determined and desired speeds of the motor **222**. For instance, the magnitude of the current adjustment may be based on a present difference, a past difference (based on accumulated error over time), and a future difference (based on current rates of change) between the determined and desired speeds of the spool.

In some embodiments, the tether control module **216** may vary the rate at which the tether **224** and payload **228** are lowered to the ground. For example, the speed controller may change the desired operating rate according to a variable deployment-rate profile and/or in response to other factors in order to change the rate at which the payload **228** descends toward the ground. To do so, the tether control module **216** may adjust an amount of braking or an amount of friction that is applied to the tether **224**. For example, to vary the tether deployment rate, the UAV **200** may include friction pads that can apply a variable amount of pressure to the tether **224**. As another example, the UAV **200** can include a motorized braking system that varies the rate at which the spool lets out the tether **224**. Such a braking system may take the form of an electromechanical system in which the motor **222** operates to slow the rate at which the spool lets out the tether **224**. Further, the motor **222** may vary the amount by which it adjusts the speed (e.g., the RPM) of the spool, and thus may vary the deployment rate of the tether **224**. Other examples are also possible.

In some embodiments, the tether control module **216** may be configured to limit the motor current supplied to the motor **222** to a maximum value. With such a limit placed on the motor current, there may be situations where the motor **222** cannot operate at the desired operation specified by the speed controller. For instance, as discussed in more detail below, there may be situations where the speed controller specifies a desired operating rate at which the motor **222** should retract the tether **224** toward the UAV **200**, but the motor current may be limited such that a large enough downward force on the tether **224** would counteract the retracting force of the motor **222** and cause the tether **224** to unwind instead. And as further discussed below, a limit on the motor current may be imposed and/or altered depending on an operational state of the UAV **200**.

In some embodiments, the tether control module **216** may be configured to determine a status of the tether **224** and/or the payload **228** based on the amount of current supplied to the motor **222**. For instance, if a downward force is applied to the tether **224** (e.g., if the payload **228** is attached to the tether **224** or if the tether **224** gets snagged on an object when retracting toward the UAV **200**), the tether control module **216** may need to increase the motor current in order to cause the determined rotational speed of the motor **222** and/or spool to match the desired speed. Similarly, when the downward force is removed from the tether **224** (e.g., upon delivery of the payload **228** or removal of a tether snag), the tether control module **216** may need to decrease the motor current in order to cause the determined rotational speed of the motor **222** and/or spool to match the desired speed. As such, the tether control module **216** may be configured to monitor the current supplied to the motor **222**. For instance, the tether control module **216** could determine the motor current based on sensor data received from a current sensor of the motor or a current sensor of the power system **220**. In any case, based on the current supplied to the motor **222**, determine if the payload **228** is attached to the tether **224**, if someone or something is pulling on the tether **224**, and/or if the payload retriever **226** is pressing against the UAV **200** after retracting the tether **224**. Other examples are possible as well.

During delivery of the payload **228**, the payload retriever **226** can be configured to secure the payload **228** while being lowered from the UAV by the tether **224**, and can be further configured to release the payload **228** upon reaching ground level. The payload retriever **226** can then be retracted to the UAV by reeling in the tether **224** using the motor **222**.

In some implementations, the payload **228** may be passively released once it is lowered to the ground. For example, a passive release mechanism may include one or more swing arms adapted to retract into and extend from a housing. An extended swing arm may form a hook on which the payload **228** may be attached. Upon lowering the release mechanism and the payload **228** to the ground via a tether, a gravitational force as well as a downward inertial force on the release mechanism may cause the payload **228** to detach from the hook allowing the release mechanism to be raised upwards toward the UAV. The release mechanism may further include a spring mechanism that biases the swing arm to retract into the housing when there are no other external forces on the swing arm. For instance, a spring may exert a force on the swing arm that pushes or pulls the swing arm toward the housing such that the swing arm retracts into the housing once the weight of the payload **228** no longer forces the swing arm to extend from the housing. Retracting the swing arm into the housing may reduce the likelihood of the release mechanism snagging the payload **228** or other nearby objects when raising the release mechanism toward the UAV upon delivery of the payload **228**.

Active payload release mechanisms are also possible. For example, sensors such as a barometric pressure based altimeter and/or accelerometers may help to detect the position of the release mechanism (and the payload) relative to the ground. Data from the sensors can be communicated back to the UAV and/or a control system over a wireless link and used to help in determining when the release mechanism has reached ground level (e.g., by detecting a measurement with the accelerometer that is characteristic of ground impact). In other examples, the UAV may determine that the payload has reached the ground based on a weight sensor detecting a threshold low downward force on the tether and/or based on a threshold low measurement of power drawn by the winch when lowering the payload.

Other systems and techniques for delivering a payload, in addition or in the alternative to a tethered delivery system are also possible. For example, a UAV **200** could include an air-bag drop system or a parachute drop system. Alternatively, a UAV **200** carrying a payload could simply land on the ground at a delivery location. Other examples are also possible.

IV. ILLUSTRATIVE UAV DEPLOYMENT SYSTEMS

UAV systems may be implemented in order to provide various UAV-related services. In particular, UAVs may be provided at a number of different launch sites that may be in communication with regional and/or central control systems. Such a distributed UAV system may allow UAVs to be quickly deployed to provide services across a large geographic area (e.g., that is much larger than the flight range of any single UAV). For example, UAVs capable of carrying payloads may be distributed at a number of launch sites across a large geographic area (possibly even throughout an entire country, or even worldwide), in order to provide on-demand transport of various items to locations throughout the geographic area. FIG. **3** is a simplified block diagram illustrating a distributed UAV system **300**, according to an example embodiment.

In the illustrative UAV system **300**, an access system **302** may allow for interaction with, control of, and/or utilization of a network of UAVs **304**. In some embodiments, an access system **302** may be a computing system that allows for human-controlled dispatch of UAVs **304**. As such, the control system may include or otherwise provide a user interface through which a user can access and/or control the UAVs **304**.

In some embodiments, dispatch of the UAVs **304** may additionally or alternatively be accomplished via one or more automated processes. For instance, the access system **302** may dispatch one of the UAVs **304** to transport a payload to a target location, and the UAV may autonomously navigate to the target location by utilizing various on-board sensors, such as a GPS receiver and/or other various navigational sensors.

Further, the access system **302** may provide for remote operation of a UAV. For instance, the access system **302** may allow an operator to control the flight of a UAV via its user interface. As a specific example, an operator may use the access system **302** to dispatch a UAV **304** to a target location. The UAV **304** may then autonomously navigate to the general area of the target location. At this point, the operator may use the access system **302** to take control of the UAV **304** and navigate the UAV to the target location (e.g., to a particular person to whom a payload is being transported). Other examples of remote operation of a UAV are also possible.

In an illustrative embodiment, the UAVs **304** may take various forms. For example, each of the UAVs **304** may be a UAV such as those illustrated in FIGS. **1**A-**1**E. However, UAV system **300** may also utilize other types of UAVs without departing from the scope of the invention. In some implementations, all of the UAVs **304** may be of the same or a similar configuration. However, in other implementations, the UAVs **304** may include a number of different types of UAVs. For instance, the UAVs **304** may include a number of types of UAVs, with each type of UAV being configured for a different type or types of payload delivery capabilities.

The UAV system **300** may further include a remote device **306**, which may take various forms. Generally, the remote device **306** may be any device through which a direct or indirect request to dispatch a UAV can be made. (Note that an indirect request may involve any communication that may be responded to by dispatching a UAV, such as requesting a package delivery). In an example embodiment, the remote device **306** may be a mobile phone, tablet computer, laptop computer, personal computer, or any network-connected computing device. Further, in some instances, the remote device **306** may not be a computing device. As an example, a standard telephone, which allows for communication via plain old telephone service (POTS), may serve as the remote device **306**. Other types of remote devices are also possible.

Further, the remote device **306** may be configured to communicate with access system **302** via one or more types of communication network(s) **308**. For example, the remote device **306** may communicate with the access system **302** (or a human operator of the access system **302**) by communicating over a POTS network, a cellular network, and/or a data network such as the Internet. Other types of networks may also be utilized.

In some embodiments, the remote device **306** may be configured to allow a user to request delivery of one or more items to a desired location. For example, a user could request UAV delivery of a package to their home via their mobile phone, tablet, or laptop. As another example, a user could request dynamic delivery to wherever they are located at the time of delivery. To provide such dynamic delivery, the UAV system **300** may receive location information (e.g., GPS coordinates, etc.) from the user's mobile phone, or any other device on the user's person, such that a UAV can navigate to the user's location (as indicated by their mobile phone).

In an illustrative arrangement, the central dispatch system **310** may be a server or group of servers, which is configured to receive dispatch messages requests and/or dispatch instructions from the access system **302**. Such dispatch messages may request or instruct the central dispatch system **310** to coordinate the deployment of UA Vs to various target locations. The central dispatch system **310** may be further configured to route such requests or instructions to one or more local dispatch systems **312**. To provide such functionality, the central dispatch system **310** may communicate with the access system **302** via a data network, such as the Internet or a private network that is established for communications between access systems and automated dispatch systems.

In the illustrated configuration, the central dispatch system **310** may be configured to coordinate the dispatch of UAVs **304** from a number of different local dispatch systems **312**. As such, the central dispatch system **310** may keep track of which UAVs **304** are located at which local dispatch systems **312**, which UAVs **304** are currently available for deployment, and/or which services or operations each of the UAVs **304** is configured for (in the event that a UAV fleet includes multiple types of UAVs configured for different services and/or operations). Additionally or alternatively, each local dispatch system **312** may be configured to track which of its associated UAVs **304** are currently available for deployment and/or are currently in the midst of item transport.

In some cases, when the central dispatch system **310** receives a request for UAV-related service (e.g., transport of an item) from the access system **302**, the central dispatch system **310** may select a specific UAV **304** to dispatch. The central dispatch system **310** may accordingly instruct the local dispatch system **312** that is associated with the selected UAV to dispatch the selected UAV. The local dispatch system **312** may then operate its associated deployment system **314** to launch the selected UAV. In other cases, the central dispatch system **310** may forward a request for a UAV-related service to a local dispatch system **312** that is near the location where the support is requested and leave the selection of a particular UAV **304** to the local dispatch system **312**.

In an example configuration, the local dispatch system **312** may be implemented as a computing system at the same location as the deployment system(s) **314** that it controls. For example, the local dispatch system **312** may be implemented by a computing system installed at a building, such as a warehouse, where the deployment system(s) **314** and UAV(s) **304** that are associated with the particular local dispatch system **312** are also located. In other embodiments, the local dispatch system **312** may be implemented at a location that is remote to its associated deployment system(s) **314** and UAV(s) **304**.

Numerous variations on and alternatives to the illustrated configuration of the UAV system **300** are possible. For example, in some embodiments, a user of the remote device **306** could request delivery of a package directly from the central dispatch system **310**. To do so, an application may be implemented on the remote device **306** that allows the user to provide information regarding a requested delivery, and generate and send a data message to request that the UAV system **300** provide the delivery. In such an embodiment, the central dispatch system **310** may include automated functionality to handle requests that are generated by such an application, evaluate such requests, and, if appropriate, coordinate with an appropriate local dispatch system **312** to deploy a UAV.

Further, some or all of the functionality that is attributed herein to the central dispatch system **310**, the local dispatch system(s) **312**, the access system **302**, and/or the deployment system(s) **314** may be combined in a single system, implemented in a more complex system, and/or redistributed among the central dispatch system **310**, the local dispatch system(s) **312**, the access system **302**, and/or the deployment system(s) **314** in various ways.

Yet further, while each local dispatch system **312** is shown as having two associated deployment systems **314**, a given local dispatch system **312** may alternatively have more or fewer associated deployment systems **314**. Similarly, while the central dispatch system **310** is shown as being in communication with two local dispatch systems **312**, the central dispatch system **310** may alternatively be in communication with more or fewer local dispatch systems **312**.

In a further aspect, the deployment systems **314** may take various forms. In general, the deployment systems **314** may take the form of or include systems for physically launching one or more of the UAVs **304**. Such launch systems may include features that provide for an automated UAV launch and/or features that allow for a human-assisted UAV launch. Further, the deployment systems **314** may each be configured to launch one particular UAV **304**, or to launch multiple UAVs **304**.

The deployment systems **314** may further be configured to provide additional functions, including for example, diagnostic-related functions such as verifying system functionality of the UAV, verifying functionality of devices that are housed within a UAV (e.g., a payload delivery apparatus), and/or maintaining devices or other items that are housed in the UAV (e.g., by monitoring a status of a payload such as its temperature, weight, etc.).

In some embodiments, the deployment systems **314** and their corresponding UAVs **304** (and possibly associated local dispatch systems **312**) may be strategically distributed throughout an area such as a city. For example, the deployment systems **314** may be strategically distributed such that each deployment system **314** is proximate to one or more payload pickup locations (e.g., near a restaurant, store, or warehouse). However, the deployment systems **314** (and possibly the local dispatch systems **312**) may be distributed in other ways, depending upon the particular implementation. As an additional example, kiosks that allow users to transport packages via UAVs may be installed in various locations. Such kiosks may include UAV launch systems, and may allow a user to provide their package for loading onto a UAV and pay for UAV shipping services, among other possibilities. Other examples are also possible.

In a further aspect, the UAV system **300** may include or have access to a user-account database **316**. The user-account database **316** may include data for a number of user accounts, and which are each associated with one or more persons. For a given user account, the user-account database **316** may include data related to or useful in providing UAV-related services. Typically, the user data associated with each user account is optionally provided by an associated user and/or is collected with the associated user's permission.

Further, in some embodiments, a person may be required to register for a user account with the UAV system **300**, if they wish to be provided with UAV-related services by the UAVs **304** from UAV system **300**. As such, the user-account database **316** may include authorization information for a given user account (e.g., a username and password), and/or other information that may be used to authorize access to a user account.

In some embodiments, a person may associate one or more of their devices with their user account, such that they can access the services of UAV system **300**. For example, when a person uses an associated mobile phone, e.g., to place a call to an operator of the access system **302** or send a message requesting a UAV-related service to a dispatch system, the phone may be identified via a unique device identification number, and the call or message may then be attributed to the associated user account. Other examples are also possible.

V. EXAMPLE OPERATION OF CONTEXT-BASED NAVIGATION OF UAVS USING RELATIVE-POSITION PATTERNS OF FIDUCIAL MARKERS

Context-based navigation of UAVs using patterns or constellations of fiducial markers may be described by way of example operation by considering a UAV flying to a designated target charging pad among a cluster of charging pads. In an example usage scenario, such a UAV may have a mission that includes (among possibly other actions) flying to and landing on the designated target charging pad. The example operation, described below, is not intended to be limiting with respect to the scope of example embodiments described herein.

The UAV system **300** illustrated in FIG. **3** is an example of a support infrastructure described briefly above in connection with battery charging pads and mission and operations support. In connection with these and other aspects of UAV fleet operation, the deployment system(s) **314** of FIG. **3** may be considered a jet bridge. More particularly, the deployment system(s) **314** may function as launching and landing sites or platforms for UAVs, and, as shown, may also be communicatively connected with other components of the infrastructure, including those which may support mission operations and data processing/analysis. As such, a deployment system **314** may provide a variety of support operations for UAVs, including communications with network-based servers and other infrastructure elements. Deployment system(s) **314** may also include clusters of charging pads.

Although not necessarily shown as distinct infrastructure entities in FIG. **3**, charging pad clusters may also be deployed as standalone configurations. In accordance with example embodiments, a standalone charging pad cluster may include a layout of charging pads connected to AC power, and possibly configured to function as little more than landing pads or platforms with battery charging facilities. In practice, charging pad clusters may be located at delivery pick-up sites, such as vendors' warehouses or retail stores, for example. UAVs may be controlled and/or preprogrammed to visit such sites to retrieve items for delivery, and to charge their batteries during their stopovers, but may not necessarily need any other services during each such stop.

FIG. **4**A illustrates an example charging pad cluster **400**, in accordance with example embodiments. In the illustration, charging pads **402** are represented by squares at regular positions in a square grid (or array) having five charging pads on each side for a total of 25 charging pads. As mentioned above and discussed in more detail below, a charging pad cluster may also include a plurality of fiducial markers at positions distributed across the layout (e.g., grid) of charging pads. For the moment, fiducial markers are omitted from the depiction of charging pad cluster **400** in FIG. **4**A. However, a distribution of fiducial markers will be depicted in a subsequent figure (see FIG. **4**D) and discussed in connection therewith.

Continuing with FIG. **4**A, axes labeled "X" and "Y" at the lower left of the grid represent a coordinate system for the grid. By way of example, all the charging pads are oriented with their sides parallel to the axes. For purposes of the discussion here, two of the charging pads are distinctly labeled **402**-A and **402**-B. Also by way of example, all the charging pads **402** are identical in size, having dimensions of D_x and D_y, as shown. As also shown, the center-to-center separation of the charging pads is S_x in the X-direction and S_y in the Y-direction. And again by way of example, the X and Y separations are equal ($S_x=S_x$). In the discussion of example operation below, charging pad **402**-A serves as an example designated target charging pad for a UAV, so it is depicted with a thick black line as a visual cue of its special role in the discussion.

It should be understood that other configurations of charging pads in a cluster are possible as well. For example, the layout need not necessarily be square or even regular, and the orientation of the charging pads need not necessarily be identical or aligned with the sides of a grid. Further, a cluster need not necessarily include only identically-sized charging pads. For the purposes of the present discussion, however, there is no loss in generality by considering a cluster of identically-sized, square (or rectangular) charging pads configured at regular positions in a square (or rectangular) grid, and all oriented parallel to the sides of the grid. Non-limiting examples of dimensions of an example embodiment of charging pad cluster **400** include $D_x=D_y=1.2$ meters (4 feet), and $S_x=S_y=3$ meters (9.8 feet). Note that the charging pad cluster **400** in FIG. **4**A is not necessarily to scale.

For purposes of the discussion herein, the vicinity of a charging pad cluster will be referred to as its "locale." Non-limiting examples of a locale of charging cluster may be a warehouse in which, or adjacent to which, the charging pad cluster is located. Other terms similar to "locale" may also be used to convey its meaning as applied herein, such as "locality," "region," and "site," among others.

In an example operational scenario such as the one described above, a UAV may intend or be instructed (e.g., according to a mission flight plan) to fly to and land on a designated target charging pad—for example, charging pad **402**-A of charging cluster **400**. The specification of a designated target charging pad from among a cluster may be related to particular operational protocols of a fleet of UAVs. For example, coordination of multiple flight plans of multiple UAVs of a fleet may call for or require advanced assignments of designated charging pads for landing. There could be other or additional reasons as well.

In an example embodiment, the UAV may be configured for both flying predominantly horizontally between different geolocations, and ascending, descending, and hovering vertically over any particular location on the ground. For purposes of the discussion herein, it may be assumed that in flying from ground-point "A" to ground-point "B," a UAV may ascend vertically from ground-point A, fly predominantly horizontally to a point vertically above ground-point B, then descend to land at ground-point B. Further, the UAV may need to confirm that it has correctly identified ground-point B before descending to land there, and the UAV may possibly adjust its horizontal position if necessary first. However, it should be understood that while the vertical ascent, descent, and hovering operations may be useful and/or helpful in describing and illustrating example operation, they are not necessarily required aspects of example embodiments. In particular, the example operation described below may be adapted to accommodate UAVs that take off and land using runways, similarly to commercial airplanes, for example.

The UAV, such as UAV **200**, may have a navigation system, as described above, and may use the navigation system when flying to the locale of charging cluster **400**. Although not necessarily shown in FIG. **2** or explicitly described in the associated discussion, the navigation system may be configured to operate in more than one mode. In particular, and in accordance with example embodiments, the navigation system may be configured to operate in at least a "coarse-navigation" mode and a "context-based fine-navigation mode," or just "context-based" navigation mode.

In accordance with example embodiments, the coarse-navigation mode may enable the UAV to fly to location that is within a tolerance of a desired or designated location, so as to arrive at a "coarse-navigation" position or location. The coarse-navigation position may be vertically above an approximate ground position of the designated target charging pad, but insufficiently accurate for the UAV to distinguish its designated target charging pad from one or more other nearby (e.g., neighboring) charging pads. The UAV may then use (e.g., switch to) the context-based navigation mode, which enables the UAV to distinguish its designated target charging pad from the other nearby charging pads based on a comparison of observed constellations of fiducial markers relative to the charging pads with known reference constellations in a reference map of the charging pad cluster **400**. The UAV may then descend and land on its designated target charging pad once it confirms it has correctly identified its designated pad.

In an example, the coarse-navigation mode may use a GPS system. Such a system may be subject to limitations of accuracy due to inherent operational factors and/or or environmental factors that may degrade or diminish a received GPS signal, for example. Inherent limitations on accuracy could correspond to best-case accuracy or accuracy under best practical environmental and/or operational conditions. Environmental factors could be dependent on the locale of the charging cluster or particular locations within the locale, for example. In practice, environmental factors could affect the likelihood that a given degree of inherent accuracy will be achieved on any given flight. However, the likelihood could also include random factors as well. In any case, in using the coarse-navigation mode to fly to the locale, the UAV may thus arrive to within the tolerance of charging pad **402**-A with a given likelihood. This may be expressed as the UAV having a "Q" percent chance of arriving to within "R" meters of a position directly above the designated target charging pad.

FIG. **4**B is a conceptual illustration of a coarse-navigation position with respect to charging pad **402**-A of charging pad cluster **400**, in accordance with example embodiments. In this illustration, a few different tolerances ("R" in the expression above) of distance from a center point of charging pad **402**-A as viewed from above are represented as concentric rings of grayscale, and the associated likelihoods ("Q" in the expression above) are indicated as percentages within the tolerances. Taking an example in which R is normally distributed, Q=99% could be interpreted as a 3σ ("3-sigma") statistic. An example 30 tolerance is indicated by a dashed-dotted circle labeled "Coarse-Navigation Accuracy." For illustrative purposes, and by way of example, the 30 tolerance is taken to be approximately 3 meters. That is, a UAV using its coarse-navigation mode to fly to charging pad **402**-A may be expected to arrive within 3 meters of a position above charging pad **402**-A with a 99% likelihood. In this example illustration, it also appears that the UAV may be expected to arrive within approximately 1.5, 1.0, and 0.5 meters with approximately 93%, 84%, and 70% likelihoods, respectively. These likelihoods simply express integrals of a normal distribution having σ=1 meter. Note that the concentric rings of grayscale in FIG. **4**B are not necessarily to scale.

It should be understood that the example tolerances and likelihoods above are largely hypothetical, and presented for illustrative purposes. Similarly, the assumption of a normal distribution of coarse-navigation arrival locations is also presented by way of example. In some implementations, the actual values and distributions could be different, though the example values may not be unreasonable estimates of actual values that may be found in practice. Putting aside the hypothetical nature of the tolerances and likelihoods above, FIG. **4**B serves to illustrate that there is a degree of approximation of the coarse-navigation position and a resulting potential for mis-identification of a designated target charging pad.

Operationally, when a UAV arrives at, or as the UAV is approaching, a charging pad cluster, the UAV may initially identify a charging pad closest to a point below its coarse-navigation arrival position as a "best guess" (or "candidate") for the designated target charging pad. FIG. **4**B also depicts a dashed-dotted circle at 1.5 meters, with a diagonal arrow labeled "~7%" indicating a likelihood that a coarse-navigation arrival position will be at or beyond this circle. Since 1.5 meters marks roughly a half-way point between charging pad **402**-A and its immediate neighboring charging pads (e.g., charging pad **402**-B), a UAV arriving at a coarse-navigation position beyond the 1.5 meters circle may be subject to mis-identifying a neighboring charging pad as charging pad **402**-B, if identification is based solely on the closest charging pad to the UAV's coarse-navigation arrival position. The questions marks ("?") in the neighboring charging pads of charging pad **402**-A signify this potential ambiguity in identification. For the current example, and without any additional procedure for disambiguation, such a mis-identification may be expected to occur (or at least may be at risk of occurring) with an approximate likelihood of 7%, or approximately 7% of the time. Again, this value and the assumptions that lead to it are presented by way of example. However, the potential mis-identification by a UAV of a designated target charging pad due to limitations on the accuracy of the coarse-navigation mode applies at least conceptually to these and other possible statistical descriptions of navigational accuracy.

In order to aid or achieve disambiguation of identities of charging pads of a charging pad cluster, fiducial markers may be distributed among the charging pads to serve as distinctive visual cues to an imaging system of a UAV. One approach to deploying fiducial markers for this purpose is to provide a set of fiducial markers, each of which is uniquely visually identifiable from among the set, and each of which is placed at a precise (or nearly so) geolocation within a layout of a charging pad cluster. For example, each charging pad may have one or more uniquely identifiable fiducial markers associated with it. The UAV may be provided with information, such as a data table, that lists each charging pad and its associated fiducial markers. The UAV may then be able to cross-check observed fiducial marker identifications and charging pads against the data table in order to resolve mis-identifications of charging pads that may arise due to imprecise navigation, for example. In addition to the use of uniquely identifiable fiducial markers, and precisely (or nearly so) determining each of their geolocations (possibly as well as a specific orientation for each), this approach may also call for relatively regular or routine site maintenance in order to help ensure an accurate and/or up-to-date match between the site deployment and the data table that records it.

Example embodiments herein provide an alternative methodology for using fiducial markers to aid and/or achieve disambiguation of identities of charging pads of a charging pad cluster. Specifically, the alternative methodology involves using patterns or constellations of fiducial markers formed by their relative positions and orientations with respect to charging pads of a cluster to distinguish between the charging pads. In accordance with example embodiments, a respective sub-plurality of fiducial markers in the vicinity of each respective charging pad may form a different apparent constellation in relation to the respective charging pad. A reference map of the cluster that captures an aggregate of the respective constellations may be generated and provided to a UAV, and may then be used by the UAV for comparing constellations observed with its imaging system with "reference constellations" in the reference map. The UAV may thus use the comparisons to disambiguate between observed charging pads, and thereby correctly identify a designated target charging pad.

FIG. **4**C illustrates an example charging pad cluster **400** with example fiducial markers **404**, in accordance with example embodiments. For purposes of discussion, the fiducial markers **404** are represented somewhat cartoonishly as arbitrary shapes and with graphic patterns that serve as identifiable features. Thus, as shown the orientation of a fiducial may be distinguishable by its shape (e.g., having an asymmetry and/or a graphic pattern, such as graphic markers or numbers). The positions of the fiducial markers **404** across a layout of charging pads of a cluster may be configured according to a specific pattern, randomly assigned, or some blend of both, for example. Any arrangement should yield distinguishable constellations among the charging pads.

As shown in FIG. **4**C, there is some repetition among the fiducial markers **404**. In the example illustrated, instances of identical fiducial markers appear in different orientations, but this need not be a requirement, so long as distinguishable constellations can emerge from the distribution of fiducial markers. However, using different orientations among identical fiducial markers may enhance the recognizability of constellations. In any case, the ability to distinguish different constellations even with the inclusion of at least some identical fiducial markers may make it possible to use at least some non-uniquely identifiable fiducial markers among a plurality of fiducial markers deployed across a charging pad cluster.

In some embodiments, the fiducial markers may be "April tags." As is known, April tags comprise graphical patterns of "low resolution" QR-like codes. Three square fiducial markers in FIG. **4**C represent an example of the same April tag in three different orientations, distinguishable by the orientation of the graphic pattern and/or of the square shape. For at least the purpose of this discussion, and possibly in at least one or more actual deployments, however, fiducial markers in addition to, or other than, April tags may be used.

The illustrative depiction of charging pad cluster **400**, including the fiducial markers, in FIG. **4**C may be considered as representing a specified layout of charging pads **402** and fiducial markers **404**, or as representing a view or image taken from above of an actual deployment. In the latter case, an aerial image of charging pad cluster **400** may be captured by the imaging system of a UAV that is hovering directly over charging pad **402**-A at the center of the grid, for example. Such an image may be used to generate a graphical reference map of the charging pad cluster **400**, which may then be provided to UAVs having missions that include flying to and landing on a designated target charging pad of the cluster.

In accordance with example embodiments, a server in an infrastructure support network for UAVs may receive one or more aerial images of charging pad cluster **400**, and may process the one or more aerial images to generate a graphical reference map of the charging pad cluster **400**. The reference map may represent an optimization of the one or more images in a form that may correct or compensate for possibly multiple perspectives and positions with respect to the charging pad cluster **400** from which the one or more images may have be captured. One or more UAVs may have captured the one or more images, and then transmitted (or otherwise provided) them to the server. The one or more UAVs may have captured the one or more images as part of specific missions to survey the charging pad cluster for the purpose of obtaining the images for generating the reference map. Additionally or alternatively, the one or more UAVs may have flown to the charging pad cluster **400** and captured the images as part of operations to disambiguate charging pads in order to identify a designated target charging pad. Such images may be used by the server to update, revise, and/or further optimize an existing reference map.

In some example operations, a UAV may start a mission from a particular "launching" charging pad. In an initial operational state prior to flight, the UAV may not have awareness of the charging pad identity, and may first ascend vertically to obtain an aerial image of the charging pad cluster in order to apply context-based navigation to determine its launching charging pad. The aerial image so obtained may be sent from the UAV to the server to provide a further image for updating and/or optimizing an existing reference map.

FIG. **4**D illustrates an example charging pad cluster with example fiducial markers as viewed from two different hypothetical perspectives, in accordance with example embodiments. Two panels are shown, panel (a) above and panel (b) below, each conceptually representing an aerial image of charging pad cluster **400** as captured from a position above a different charging pad. In each panel, an "X" marks the charging pad above which the image was (hypothetically) obtained from. The perspectives as illustrated are somewhat exaggerated, and not necessarily intended to be to scale or otherwise accurate renditions of how the charging pad cluster **400** would actually appear from the positions above the marked charging pads.

In accordance with example embodiments, the server may receive multiple such aerial images from the UAV(s) that obtained them. The UAV(s) may also provide information associated with each image (e.g., metadata) that can be used by the server to correct and/or compensate for the particular perspective of the image. Such information may include the height of the UAV above the charging pad from which the image was obtained, and the orientation (e.g., yaw) of the UAV when the image was obtained. Other information may be included as well. The process of generating an optimized image may use the images and information to produce a sort of synthesized graphical map or image that may depict, with reasonable accuracy, how any given charging pad and its vicinity on the ground would appear from directly above the given charging pad. In particular, the reference map may be used as a sort of virtual rendition, as viewed from above, of the constellation of fiducial markers in the vicinity of any given charging pad of the cluster.

FIG. **5** is a representation of a reference map **500** of charging pad cluster **400**, also depicting of two portions of the map that represent reference constellations **502**-A and **502**-B of neighboring charging pads **402**-A and **402**-B, respectively. In accordance with example embodiments, the reference map **500** may take the form of data that may be rendered graphically and/or computationally interpreted in a form equivalent or analogous to a graphical rendering. In FIG. **5**, the map **500**, shown on the right-hand side, is presented on a scroll-like background as a visual cue for the discussion that the figure illustrates a rendering of data. In some examples, the rendering could take the form of a synthesized image, for example. The reference constellations **502**-A and **502**-B, shown to the left of the map and similarly presented on scroll-like backgrounds, could be generated distinct subsets of the data of the reference map **500**, or identified sub-regions of the reference map but not separately generated. In either case, they serve to illustrate what an idealized view from directly above the respective charging pads **402**-A and **402**-B may be expected to look like.

More particularly, reference constellation **502**-A represents a constellation of fiducial markers at relative positions and orientations with respect to charging pad **402**-A as that constellation would be expected to ideally appear from a position above charging pad **402**-A. Similarly, reference constellation **502**-B represents a constellation of fiducial markers at relative positions and orientations with respect to charging pad **402**-B as that constellation would be expected to ideally appear from a position above charging pad **402**-B. For the purposes of clarity in the figure, each reference constellation omits the neighboring charging pads. This omission could also be an intentional feature of some example embodiments that generate reference constellations separately from a reference map from which they are derived.

The qualification of a reference view or appearance as "ideal" may be considered a recognition that an actual aerial view or image from a UAV positioned above a charging pad and its associated constellation of nearby fiducial markers may differ to some degree from what appears in the reference map **500** or the reference constellations. Differences between an observed view or image from above an actual charging pad and its appearance a reference map or reference constellation may be due to various factors. Examples of such factors may include differences in the UAV's orientation and/or altitude with respect to a charging pad cluster compared with an idealized orientation of the reference map; a missing, damaged, or obscured fiducial marker at the charging pad cluster that has not been accounted or corrected for in the reference map, and possible distortions of the idealize view of the reference map due to imperfect optimization of the aerial images used to generate the map. However, in accordance with example embodiments, the spatial distribution of fiducial markers across an arrangement (e.g., grid) of charging pads of a charging pad cluster may be configured such that alignment or matching of an observed constellation to a reference constellation need not necessarily be exact in order to achieve a threshold degree of agreement that may be interpreted as a confirmed match. In this sense, a confirmed match may tolerate some amount of disagreement.

FIG. **6** is a conceptual illustration of an example operation for disambiguating charging pad identities using context-based navigation, in accordance with example embodiments. Continuing from the discussion above, it may be assumed that charging pad **402**-A is a designated target charging pad for a UAV having a mission that includes flying to the charging pad cluster **400**, and as described, the UAV may arrive initially at a coarse-navigation location that is above a ground point within approximately 3 meters of the designated target charging pad. For any particular instance, this arrival location could be closer to one of the neighboring charging pads than to charging pad **402**-A. FIG. **6** illustrates an example of how the UAV may use the reference constellations **502**-A and **502**-B to distinguish between charging pad **402**-A and **402**-B. The need to do so could arise if the UAV's coarse arrival location were above a ground point halfway between the two charging pads, or otherwise insufficiently close to either one to make a confident identification. Additionally or alternatively, UAVs could apply the disambiguation operation routinely, even if their coarse-navigation arrival locations are directly above a charging pad since, as discussed in connection with FIG. **4**B, the coarse-navigation location could be above the "wrong" charging pad ~7% of the time (depending on specifics of the coarse-navigation arrival location probability distribution, for example).

In accordance with example embodiments, a UAV may be provided by the server with a reference map **500** prior to arriving at a locale of charging pad cluster **400**. For example, the UAV may be provisioned with the reference map **500** as part of preparing the UAV for a mission. In another example, the reference map **500** may be transmitted to the UAV at some point after it begins its flight. If reference constellations of the charging pads are generated separately from the reference map **500** (e.g., as separate data entities, for example), the UAV may also be provided with reference constellations **502**-A and **502**-B, possibly as well as reference constellations for one or more other charging pads of the cluster. Additionally or alternatively, the UAV may have a computational capability to generate reference constellations **502**-A and **502**-B (and others) from the reference map **500**. Reference constellations **502**-A and **502**-B shown in FIG. **6** may represent the UAV's copies of these data as obtained or generated by any of the above, or possibly other, means.

Upon arriving at the locale of the charging pad cluster **400**, or shortly thereafter, the UAV may acquire one or more aerial images of one or more charging pads nearby a ground point below its coarse-navigation arrival location. The UAV may use its imaging system to obtain the one or more images. In one operational example, each image may capture a different one of the charging pads and an observed associated constellation of fiducial markers. Additionally or alternatively, more than one image may be acquired for at least one of the charging pads. As a further or alternative example, the images may correspond to different portions of a continuous video image acquired (or as it is being acquired) while the UAV hovers in place and/or moves about above the charging pad cluster **400**. As still another example, each of one or more images of one or more charging pads may be acquired from respective locations above each of the one or more charging pads. In yet another example, a single image from the UAV's coarse-navigation arrival location may suffice if the UAV's altitude is high enough to render any differences in perspective due to horizontal translations to be negligible in their effect on matching observed constellations to reference constellations.

Which one or more of the above modes of image acquisition is used may depend on operational circumstances, such that a UAV may decide which one best accommodates any particular disambiguation task instance. Additionally or alternatively, a sequence of disambiguation procedures, including what images should be acquired and how, may be more fully specified ahead of time. It should be understood that any of the above or other alternatives may accommodate, or be accommodated by, the principles of context-based navigation as described herein.

For purposes of the present discussion, only reference constellations **502**-A and **502**-B and aerial images of charging pads **402**-A (the designated target charging pad) and **402**-B are considered. In practice, the disambiguation operation may need to consider the designated target charging pad and more than one other neighboring or nearby charging pads. As represented in FIG. **6**, observed constellations **602**-A and **602**-B are captured in respective images acquired by the UAV's imaging system. The depictions of the images shown in FIG. **6** may be considered representations of separate image captures in an image plane (e.g., pixel array) of an imaging system, or different regions of a single image capture in the image plane. In some examples, the images may be still images, and in some examples they may represent frames of one or more video images. These examples are not meant to be mutually exclusive, but rather may be different options or modes of operation of the imaging system.

The images may be acquired by any one or more of the example operations described above. Each image depicted is at an angle with respect to the reference constellations **502**-A and **502**-B to signify a possible orientation of the UAV and/or the camera component of its imaging system with respect to a reference direction of the charging pad cluster **400** (e.g., as define by the X and Y axes of FIG. **4**A). For purposes of clarity in the figure, neighboring and/or nearby charging clusters are omitted from the images. In actual operation these charging pads may be visible in one or more of the images. However, their presence in an image would not be expected to change the appearance of the observed constellations.

In accordance with example embodiments, the UAV (or a controller or other processing entity thereof) may compare the observed constellations **602**-A and **602**-B of the acquired images with the reference constellations **502**-A and **502**-B in an attempt to positively identify the observed charging pad in each image. In this way, the UAV may be able to positively identify charging pad **402**-A as its designated target charging pad, and then proceed to land on it (e.g., assuming landing on the charging pad is specified in its mission). Confirmation of the identity of an observed charging pad may be carried out by a constellation-matching operation and charging pad identity disambiguation, indicated by the so-labeled block at the center of FIG. **6**. As shown by way of example, inputs to this block may be the reference constellations **502**-A and **502**-B and the UAV images of the observed constellations **602**-A and **602**-B. The output of the operation is the charging pad identity confirmation, as indicated by the so-labeled data item icon.

In making a match, the matching operation carried out by the UAV (or its controller, for example) may compensate for differences in the orientations of the acquired images and the reference constellations. In accordance with example embodiments, the matching operation may utilize, or be implemented as, a computational pattern-matching algorithm. For example, pattern matching may function by treating a charging pad and the fiducial markers of an associated constellation as vertices of a graph, and thereby seek to align vertices in an observed constellation with vertices in a reference constellation. For each attempted match, a goodness-of-fit statistic may be computed and compared with a threshold. A fit that exceeds the threshold may be considered a match. If there is more than one potential match, then one that maximally exceeds the threshold may be deemed the positive match.

In another example, the matching operation may utilize, or be implemented as, a machine-learning model or program. For example, an artificial neural network (or similar machine-learning model) may be trained to recognize matches among two or more input graphical patterns. Such a model may then be used to predict a confidence metric for a match between a given reference constellation and one or more observed constellations. The confidence metric may then be used in a similar manner to a goodness-of-fit statistic described above.

It will be appreciated that there may be other programs, algorithms, and/or procedures for determining a match between a match between a given reference constellation and one or more observed constellations. In accordance with example embodiments, matching for disambiguation of charging pad identities and/or identity confirmation may be equivalently or analogously achieved by determining a match between a given observed constellation and one or more reference constellations. A positive identification of a charging pad may be derived from either "direction" of matching.

As with the various modes of image acquisition, there can be various sequences of operations for determining a match between reference constellation **502**-A of charging pad **402**-A as it appears in the reference map **500** and one of the observed constellations in order to confirm a positive identification of the UAV's designated target charging pad. The direction of matching—i.e., matching a reference constellation to one or more observed constellations, or matching an observed constellation to one or more reference constellations—is one example of variation. Other, non-limiting examples are described in the following.

In one example, a charging pad closest to a ground point below a UAV's coarse-navigation arrival location may be selected as an initial guess for the UAV's designated target charging pad. This may be based on a known probability distribution for coarse-navigation arrival positions that peaks at a designated navigational target, as illustrated in FIG. **4**B, for example. Initially, the UAV may acquire one or more images of just the selected charging pad, and may then check if an observed constellation in the one or more images matches a reference constellation of the designated target charging pad (e.g., charging pad **402**-A in the above example). If a match found, a confirmation may be made effectively on the first try. If a match is not found, the UAV could acquire one or more images of a next-nearest charging pad, and again check if the constellation of the image(s) matches the reference constellation. If not, the process could repeat for as many observed charging pads as necessary until a match is found.

In an alternative to the above example, if the observed constellation of initial selected charging pad does not match the reference constellation of the designated target charging pad, further matching attempts against other reference constellations may be made until a match is found. This match would not identify the designated target charging pad, but would confirm an identity of the charging pad closest to the ground point above the UAV's arrival location. This information could then be used to inform or optimize a search strategy for selecting a next guess of observed charging pad to image and check for a match. Again, this process could repeat until a positive match of the UAV's designated target charging pad is determined.

In still another example, the UAV may acquire one or more images of several—e.g., four or five—charging pads nearby, and including, an initial best guess (e.g., one closest to a ground point below the UAV's arrival location). These could be obtained by the UAV moving about horizontally to locations above each of the charging pads to be imaged. Alternatively, as described above, if the altitude of the UAV's coarse-navigation arrival location is high enough, a single one (or a few) images from the arrival location may capture all of the nearby charging pads in the image field of view (FOV) with negligible perspective distortion across the image. For either image acquisition procedure, a reference constellation of the designated target charging pad could be compared against respective constellations associated with each of the charging pads in the one or more images to find out which one produces a match. The best match could then be used as a confirmation of identity of the designated target charging pad.

As still a further example, images of multiple observed constellations and multiple reference constellations may be input to the matching operation, such as a pattern matching algorithm and/or machine-learning model (e.g., trained to recognize matching patterns and output statistical confidences of detected matches). The output of the matching operation could be confirmed identities of one more of charging pads associated with the observed constellations. The UAV could then determine which of the confirmed identities corresponds to its designated target charging pad.

As noted, the above techniques and procedures for determining matches between observed constellations and reference constellations in order to disambiguate among multiple observed charging pads are not intended to be limiting. Rather, all of these and others illustrate that there may be numerous ways to implement the principle of context-based navigation, and demonstrate the flexibility and versatility of using distinguishable constellations of fiducial markers in the vicinities of observed charging pads to confirm identities of the observed charging pads. Applied by (or for) UAVs that may otherwise have only a coarse-navigation mode that provides insufficient accuracy for the UAVs to make confident identifications based on coarse-navigation arrival positions, context-based navigation thereby enables UAVs to fine tune their navigation capabilities. As also noted, a general degree of tolerance of context-based navigation to imperfect matches of observed and reference constellations may reduce and/or relax possibly stringent requirements on charging pad cluster site maintenance as it applies to upkeep of fiducial markers.

In further accordance with example embodiments, a UAV at a coarse-navigation arrival location may have a real-time communicative connection with a mission operations server in the infrastructure support network, such that some or all of the matching operations described above may be carried out by the mission operations server (or other support system of the network). As an example, the charging pad cluster at the locale of the UAV's coarse arrival location may deploy a WiFi network to which the UAV may connect, as described above, for example. The WiFi network may have a broadband interface (e.g., via a gateway router, for example) providing a link to the mission operations server. Additionally or alternatively, the UAV may have its own broadband interface (e.g., SIM card) that provides a communicative link to the mission operations server. For either of these (or other) communication scenarios, the UAV may transmit in real-time (or near real-time) some or all of its observed images to the mission operations server. The mission operations server may then carry out the analysis described above to determine a confirmed identity of the UAV's designated target charging pad, and then transmit the confirmation information back to the UAV. The UAV may the proceed based on the confirmed identity.

In some server-based implementations, the mission operations server may maintain the requisite reference maps and reference constellations, such that UAVs may not necessarily be required to carry (or be provisioned with, e.g.) their own copies. The UAVs may then only need to transmit in real-time (or near real-time) their images of observed charging pads. In other implementations, UAVs may carry (or be provided with, e.g.) reference maps and reference constellations, and may thus transmit in real-time (or near real-time) their images of observed charging pads, as well as any reference constellations needed for matching (e.g., according to one or more of the above techniques). These are just examples of context-based navigation in which a mission operations server may carry out various aspects of matching and disambiguation.

While example embodiments of context-based navigation have been described above in terms of charging pads and charging pad clusters, the principles may be applied, adapted, and/or extended to other types of navigation targets and configurations thereof. An example of a different usage scenario of context-based navigation using fiducial markers in a manner substantially similar to that described above involves automatic loading facilities used in UAV delivery operations. In an example deployment, delivery UAVs may pick up or retrieve payloads or packages from automated payload dispensers or loaders that are distinct from charging pads, and possibly configured at facilities separate from charging pad clusters. Such an operational scenario for payload pick-up may be used as an alternative to payload pick-up at charging pads, or as an additional mode that augments charging-pad pick-ups.

In accordance with example embodiments, multiple automated payload loaders, sometimes referred to as "autoloaders," may be deployed in arrangements similar to charging pads at a cluster. For example, a cluster of autoloaders may be configured in a layout at a warehouse or other storage facility. Some or all of the autoloaders may be, or appear to be, sufficiently alike as to make them potentially indistinguishable (or nearly so) to an imaging system of a UAV. A UAV with a mission plan that includes flying to a designated target autoloader at such a facility may face the same or similar navigational challenges as those described above in connection with charging pads and charging pad clusters. A plurality of fiducial markers may thus be distributed across a layout of autoloaders in the same or similar manner as that described for charging pad clusters. With such an arrangement, the principles of context-based navigating using observed and reference constellations of fiducial markers may be straightforwardly applied to clusters of autoloaders. A UAV may thus be able to distinguish between, and/or disambiguate, autoloaders, and thereby confirm an identification of a designated target autoloader.

It should be understood that confirmation of the identification of a designated target autoloader by a UAV need not necessarily be followed by the UAV landing on the target autoloader (or charging pad, for that matter). For example, in some UAV delivery scenarios, a UAV may hover above an autoloader and lower a tether to retrieve a payload or package, then reel the tether back in to secure the payload to the UAV (the same operation could apply for payload of packages retrieved from a charging pad). Other remote retrieval techniques that do not necessarily require a UAV to land may be used as well.

As another example of a different usage scenario of context-based navigation, charging pads and/or autoloaders may be equipped or configured with unique identifiers, such as unique April tags or other forms of fiducial markers, such that under expected operational conditions the charging pads and/or autoloader are visually distinguishable from one another by an imaging system of a UAV. In this arrangement, UAVs may be able to distinguish between, or disambiguate, charging pads and/or autoloaders much or most of the time—e.g., 80-90% of the time possibly on the basis of the unique markers of the charging pads alone. But the arrangement of uniquely identifiable charging pads may also serve as its own form of constellations, even without necessarily including separate fiducial markers in the manner described above. Thus, in accordance with example embodiments, context-based navigation may also be achieved as described above by using observed and reference constellations of uniquely identifiable charging pads (and/or autoloaders). As with the approach using separate fiducial markers, uniquely identifiable charging pads (and/or autoloaders) as constellations may provide the same or similar tolerance or resilience to the ability of UAVs to make positive identifications when faced with markers that become damaged, degraded, and/or obscured by detritus, for example.

These are just two examples of additional or alternative applications of context-based navigation using observed and reference constellations of fiducial markers. It should be understood that the principles may be adapted, extended, and/or applied to other usage scenarios as well.

VI. EXAMPLE METHODS

FIGS. **7** and **8** illustrate block diagrams (as flowcharts) of example methods **700** and **800** for various example embodiments described here. More particularly, example method **700**, shown in FIG. **7**, illustrates an example embodiment of context-based navigation using observed and reference constellations, as described above by way of example. Example method **800**, shown in FIG. **8**, illustrates an example embodiment of generation of a reference map, followed by use of the reference map in context-based navigation, as also described above by way of example.

A. Example UAV Method

FIG. **7** is a block diagram of method **700**, in accordance with example embodiments. In some examples, method **700** may be carried out by a control system of a UAV. In further examples, method **700** may be carried out by one or more processors, executing program instructions stored in a data storage. Execution of method **700** may involve an uncrewed vehicle, such as a UAV illustrated and described with respect to FIGS. **1**-**2**. In further examples, different blocks of method **700** may be performed by different control systems, located on and/or remote from a UAV, such as a mission operations server in an infrastructure support network for UAVs.

In accordance with example embodiments, a UAV may include a navigation system, an imaging system, and a control system. The UAV may include other systems, subsystems, and/or components as well. The control system may be configured to carry out various operations of the example method **700**.

At block **702**, method **700** includes receiving a reference map of a cluster of charging pads from a server in an infrastructure support network for UAVs. The cluster may include the charging pads arranged in a layout, and a plurality of fiducial markers distributed at positions across the layout. The reference map may represent the layout and a reference sub-plurality of the fiducial markers.

At block **704**, method **700** includes storing the reference map in a memory of the UAV.

At block **706**, method **700** includes flying to a coarse-navigation location at a locale of the cluster of charging pads.

At block **708**, method **700** includes acquiring an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads. The image may be acquired with the imaging system of the UAV. The acquired image may capture an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location.

At block **710**, method **700** includes identifying in the reference map at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map.

Finally, at block **712**, method **700** includes disambiguating an identity of a particular charging pad from among the one or more charging pads. The disambiguating may be based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence.

In accordance with example embodiments, disambiguating the identity may involve confirming that the particular charging pad is a designated landing pad for the UAV. Then, the method may further involve landing the UAV on the particular charging pad based on the confirmation.

In accordance with example embodiments, in a usage scenario where the acquired image includes two or more charging pads, disambiguating the identity may involve confirming that the particular charging pad is not a designated landing pad for the UAV. Then, the method **700** may further involve identifying a different charging pad of the two or more charging pads as the designated landing pad for the UAV, based on the known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within the degree of confidence. The UAV may then land on the different charging pad based on its identity as the designated landing pad.

In accordance with example embodiments, the layout may be a grid, and the plurality of fiducial markers may be distributed at regular positions across the grid, and/or a random positions across the grid. In addition, the fiducial markers may be identifiable by physical features that do not uniquely distinguish every fiducial marker from among the plurality. Further, at least one fiducial marker of the plurality may have a physical feature that enables detection of an angular orientation of the at least one fiducial marker.

In accordance with example embodiments, flying to the coarse-navigation location may involve using a coarse-navigation mode of the navigation system to fly the UAV, with a given likelihood of navigational success, to an arrival position that is within a coarse tolerance of a target position directly above a designated target charging pad. Then, disambiguating the identity of the particular charging pad from among the one or more charging pads may involve determining whether or not the particular charging pad is the designated target charging pad.

In accordance with example embodiments, matching the reference constellation to the observed constellation to within the degree of confidence may involve applying a pattern matching operation to the observed constellation and the reference constellation. The pattern matching operation may be implemented by a machine-learning pattern recognition program, and/or a geometric transformation computation. In an example, the geometric transformation may entail one or more rotations and/or one or more translations to align the observed and reference constellations.

In accordance with example embodiments, example method **700** may further involve sending data comprising an aerial image of the cluster of charging pads to the server. In this arrangement, the server may be configured to use the aerial image for updating a network copy of the reference map. The network copy may be provided by the server to UAVs having missions that include flying to the cluster of charging pads.

In some examples, a non-transitory computer readable medium may include program instructions executable by one or more processors to perform operations. The operations may include the steps of method **700**.

B. Example System Method

FIG. **8** is a block diagram of another method **800**, in accordance with example embodiments. Specifically, method **800** illustrates an example embodiment of generation of a reference map, followed by use of the reference map in context-based navigation. Example method **800** may be carried out by a system that includes a server in an infrastructure support network for UAVs, a cluster of charging pads for UAVs, and a UAV. The cluster of charging pads may include charging pads arranged in a layout, and a plurality of fiducial markers distributed at positions across the layout. The UAV may include a navigation system, an imaging system, and a control system configured to carry out aspects of the example method **800**.

At block **802**, method **800** includes the UAV's control system causing the UAV to fly to a locale of the cluster of charging pads.

At block **804**, method **800** includes the UAV's control system causing the imaging system of the UAV to acquire at least one aerial image of the cluster. The acquired at least one aerial image may capture an observed constellation of fiducial markers at apparent positions and orientations relative to one or more of the charging pads, as viewed from at least one position and orientation of the UAV with respect to the cluster.

At block **806**, method **800** includes the UAV sending data comprising the at least one aerial image to the server. The data may be sent as a transmission over a communicative connection that includes one or more wireless links or legs, and/or one or more wireline links or legs.

At block **808**, method **800** includes the server receiving the at least one aerial image.

At block **810**, method **800** includes the server generating and/or updating a reference map of the cluster based on the at least one aerial image.

Finally, at block **812**, method **800** includes the server providing the reference map to UAVs having missions that include flying to the cluster of charging pads.

In accordance with example embodiments, generating and/or updating the reference

map of the cluster may involve determining an optimal reference map of the cluster from one or more aerial images of the cluster. The optimal reference map may incorporate adjustments for perspectives and positions from which the one or more aerial images were acquired.

In accordance with example embodiments, example method **800** may further involve a further UAV having a further navigation system, a further imaging system, and a further control system configured to receive the reference map from the server. The further control system may also cause the further UAV to fly to a coarse-navigation location at the locale of the cluster, and cause the further imaging system to acquire an image of at least one charging pad of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the at least one charging pad. The acquired image may capture a given constellation of fiducial markers at apparent positions and orientations relative to the at least one charging pad, as viewed from a position and orientation of the further UAV at the coarse-navigation location. The further control system may identify in the reference map at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map. Then, based on known identities of the one or more reference charging pads and on matching the reference constellation to the given constellation to within a degree of confidence, the further control system may disambiguate an identity of a particular charging pad from among the at least one charging pad.

In some examples, a non-transitory computer readable medium may include program instructions executable by one or more processors to perform operations. The operations may include the steps of method **800**.

VII. CONCLUSION

The present disclosure is not to be limited in terms of the particular embodiments described in this application, which are intended as illustrations of various aspects. Many modifications and variations can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims.

The above-detailed description describes various features and functions of the disclosed systems, devices, and methods with reference to the accompanying figures. In the figures, similar symbols typically identify similar components unless context dictates otherwise. The example embodiments described herein and in the figures are not meant to be limiting. Other embodiments can be utilized, and other changes can be made without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

A block that represents a processing of information may correspond to circuitry that can be configured to perform the specific logical functions of a herein-described method or technique. Alternatively or additionally, a block that represents a processing of information may correspond to a module, a segment, or a portion of program code (including related data). The program code may include one or more instructions executable by a processor for implementing specific logical functions or actions in the method or technique. The program code or related data may be stored on any type of computer-readable medium such as a storage device including a disk or hard drive or other storage medium.

The computer-readable medium may also include non-transitory computer-readable media such as computer-readable media that stores data for short periods of time like register memory, processor cache, and random access memory (RAM). The computer-readable media may also include non-transitory computer-readable media that stores program code or data for longer periods of time, such as secondary or persistent long-term storage, like read-only memory (ROM), optical or magnetic disks, compact-disc read-only memory (CD-ROM), for example. The computer-readable media may also be any other volatile or non-volatile storage systems. A computer-readable medium may be considered a computer-readable storage medium, for example, or a tangible storage device.

Moreover, a block that represents one or more information transmissions may correspond to information transmissions between software or hardware modules in the same physical device. However, other information transmissions may be between software modules or hardware modules in different physical devices.

The particular arrangements shown in the figures should not be viewed as limiting. It should be understood that other embodiments can include more or less of each element shown in a given figure. Further, some of the illustrated elements can be combined or omitted. Yet further, an example embodiment can include elements that are not illustrated in the figures.

While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope being indicated by the following claims.

What is claimed is:

1. An uncrewed aerial vehicle (UAV), comprising:

an imaging system; and

a control system configured to:

cause the UAV to fly to a coarse-navigation location at a locale of a cluster of charging pads, the cluster comprising the charging pads arranged in a layout, and a plurality of fiducial markers distributed at positions across the layout;

cause the imaging system to acquire an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads, wherein the acquired image represents an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location;

in a reference map of the layout and a reference sub-plurality of the fiducial markers, identify at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map, wherein the reference map associates each respective charging pad of the charging pads of the cluster with a corresponding reference constellation of fiducial markers, and wherein a location of the respective charging pad within the cluster is represented (i) by respective positions and orientations of the fiducial markers of the corresponding reference constellation relative to the respective charging pad in the reference map and (ii) independently of respective geolocation data of the fiducial markers of the corresponding reference constellation; and

based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence, disambiguate an identity of a particular charging pad from among the one or more charging pads without using respective geolocation data of the reference sub-plurality of the fiducial markers.

2. The UAV of claim **1**, wherein the disambiguated identity confirms that the particular charging pad is a designated landing pad for the UAV, and wherein the control system is further configured to cause the UAV to land on the particular charging pad based on the confirmation.

3. The UAV of claim **1**, wherein the disambiguated identity confirms that the particular charging pad is not a designated landing pad for the UAV, wherein the one or more charging pads comprise two or more charging pads, and wherein the control system is further configured to:

based on the known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within the degree of confidence, identify a different charging pad of the two or more charging pads as the designated landing pad for the UAV; and

cause the UAV to land on the different charging pad based on its identity as the designated landing pad.

4. The UAV of claim **1**, further comprising a memory storing the reference map.

5. The UAV of claim **1**, wherein the layout is a grid, and wherein the plurality of fiducial markers are distributed in regular positions across the grid.

6. The UAV of claim **1**, wherein the fiducial markers are identifiable by physical features that do not uniquely distinguish every fiducial marker from among the plurality, and wherein at least one fiducial marker of the plurality has a physical feature that enables detection of an angular orientation of the at least one fiducial marker.

7. The UAV of claim **1**, wherein causing the UAV to fly to the coarse-navigation location comprises using a coarse-navigation mode of a navigation system of the UAV to fly the UAV, with a given likelihood of navigational success, to an arrival position that is within a coarse tolerance of a target position directly above a designated target charging pad, and wherein disambiguating the identity of the particular charging pad from among the one or more charging pads comprises determining whether or not the particular charging pad is the designated target charging pad.

8. The UAV of claim **1**, wherein matching the reference constellation to the observed constellation to within the degree of confidence comprises applying a pattern matching operation to the observed constellation and the reference constellation.

9. The UAV of claim **8**, wherein the pattern matching operation is implemented by a geometric transformation computation.

10. The UAV of claim **8**, wherein the pattern matching operation is implemented by a machine-learning pattern recognition program.

11. The UAV of claim **1**, wherein the control system is further configured to cause the UAV to send data comprising an aerial image of the cluster of the charging pads to a server in an infrastructure support network for UAVs, wherein the server is configured to use the aerial image for updating a network copy of the reference map, and wherein the network copy is provided by the server to UAVs having missions that include flying to the cluster of the charging pads.

12. The UAV of claim **1**, wherein the layout is a grid, and wherein the plurality of fiducial markers are distributed in random positions across the grid.

13. The UAV of claim **1**, wherein two or more fiducial markers of the plurality of fiducial markers are non-unique within the plurality of fiducial markers.

14. A method carried out by an uncrewed aerial vehicle (UAV), the method comprising:

receiving a reference map of a cluster of charging pads from a server in an infrastructure support network for UAVs, wherein the cluster comprises the charging pads arranged in a layout and a plurality of fiducial markers distributed at positions across the layout, wherein the reference map represents the layout and a reference sub-plurality of the fiducial markers, wherein the reference map associates each respective charging pad of the charging pads of the cluster with a corresponding reference constellation of fiducial markers, and wherein a location of the respective charging pad within the cluster is represented (i) by respective positions and orientations of the fiducial markers of the corresponding reference constellation relative to the respective charging pad in the reference map and (ii) independently of respective geolocation data of the fiducial markers of the corresponding reference constellation;

storing the reference map in a memory of the UAV;

flying to a coarse-navigation location at a locale of the cluster of the charging pads;

with an imaging system of the UAV, acquiring an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads, wherein the acquired image represents an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location;

in the reference map, identifying at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map; and

based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence, disambiguating an identity of a particular charging pad from among the one or more charging pads without using respective geolocation data of the reference sub-plurality of the fiducial markers.

15. The method of claim **14**, wherein disambiguating the identity comprises confirming that the particular charging pad is a designated landing pad for the UAV, and wherein the method further comprises landing the UAV on the particular charging pad based on the confirmation.

16. The method of claim **14**, wherein disambiguating the identity comprises confirming that the particular charging pad is not a designated landing pad for the UAV, wherein the one or more charging pads comprise two or more charging pads, and wherein the method further comprises:

based on the known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within the degree of confidence, identifying a different charging pad of the two or more charging pads as the designated landing pad for the UAV; and landing the UAV on the different charging pad based on its identity as the designated landing pad.

17. The method of claim **14**, wherein the layout is a grid, wherein the plurality of fiducial markers are distributed in at least one of: regular positions across the grid, or random positions across the grid, wherein the fiducial markers are identifiable by physical features that do not uniquely distinguish every fiducial marker from among the plurality, and wherein at least one fiducial marker of the plurality has a physical feature that enables detection of an angular orientation of the at least one fiducial marker.

18. The method of claim **14**, wherein matching the reference constellation to the observed constellation to within the degree of confidence comprises applying a pattern matching operation to the observed constellation and the reference constellation, and wherein the pattern matching operation is implemented by a machine-learning pattern recognition program.

19. The method of claim **14**, further comprising sending data comprising an aerial image of the cluster of the charging pads to the server, wherein the server is configured to use the aerial image for updating a network copy of the reference map, and wherein the network copy is provided by the server to UAVs having missions that include flying to the cluster of the charging pads.

20. A non-transitory computer-readable medium having stored thereon instructions that, when executed by a computing device, cause the computing device to perform operations comprising:

causing an uncrewed aerial vehicle (UAV) to fly to a coarse-navigation location at a locale of a cluster of charging pads, wherein the cluster comprises the charging pads arranged in a layout and a plurality of fiducial markers distributed at positions across the layout;

causing an imaging system of the UAV to acquire an image of one or more charging pads of the cluster and an observed sub-plurality of the fiducial markers in a vicinity of the one or more charging pads, wherein the acquired image represents an observed constellation of fiducial markers at apparent positions and orientations relative to the one or more charging pads, as viewed from a position and orientation of the UAV at the coarse-navigation location;

in a reference map of the layout and a reference sub-plurality of the fiducial markers, identifying at least a portion of a reference constellation of fiducial markers at reference positions and orientations relative to one or more reference charging pads in the reference map, wherein the reference map associates each respective charging pad of the charging pads of the cluster with a corresponding reference constellation of fiducial markers, and wherein a location of the respective charging pad within the cluster is represented (i) by respective positions and orientations of the fiducial markers of the corresponding reference constellation relative to the respective charging pad in the reference map and (ii) independently of respective geolocation data of the fiducial markers of the corresponding reference constellation; and

based on known identities of the one or more reference charging pads and on matching the reference constellation to the observed constellation to within a degree of confidence, disambiguating an identity of a particular charging pad from among the one or more charging pads without using respective geolocation data of the reference sub-plurality of the fiducial markers.

* * * * *